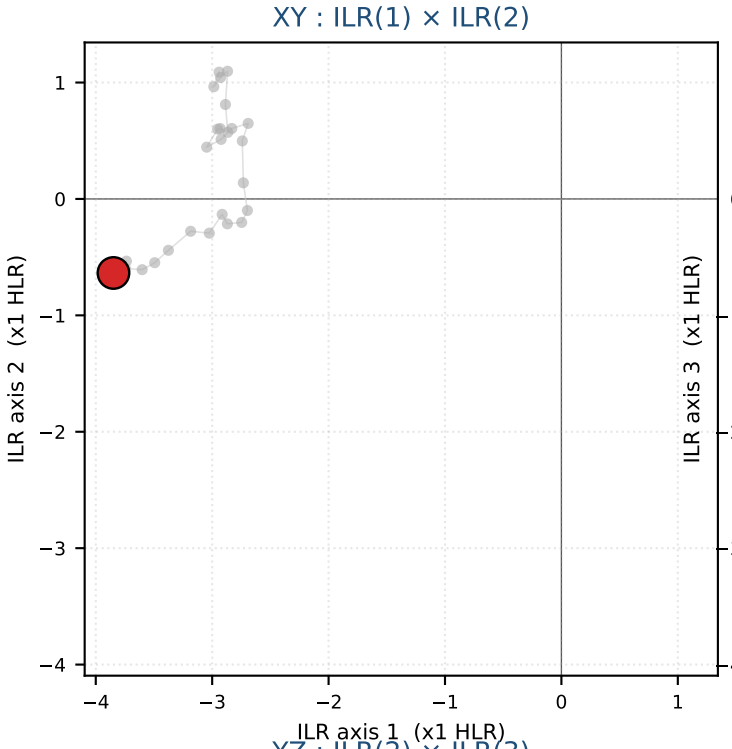
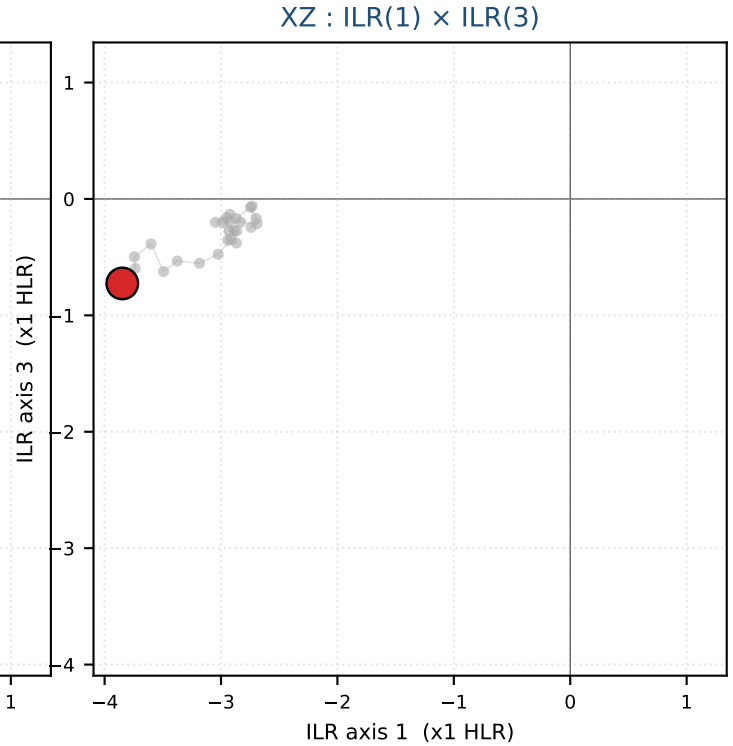
STAGE 1 v4 | t=0 | 2000 | D=8 T=26 | level 1 first-principles

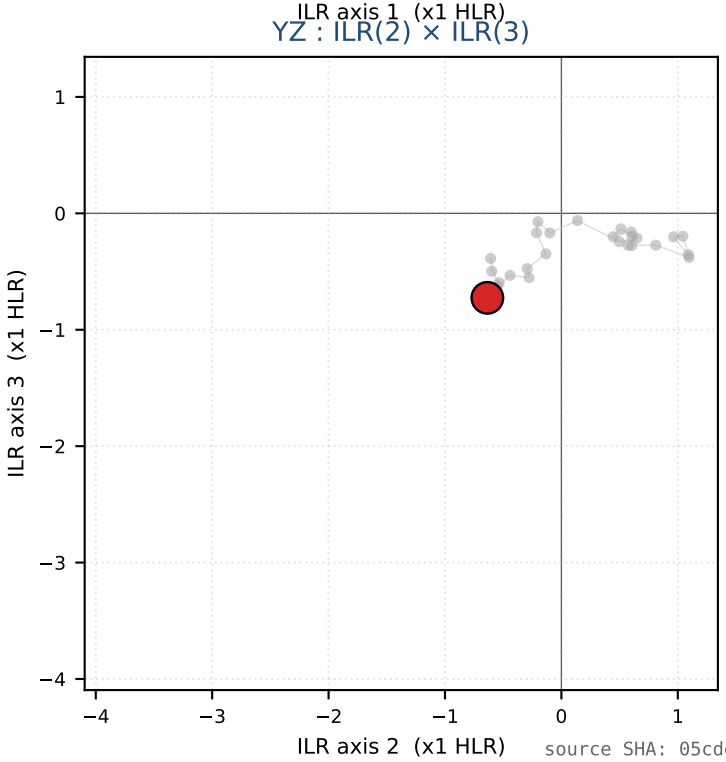
ember_ind | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

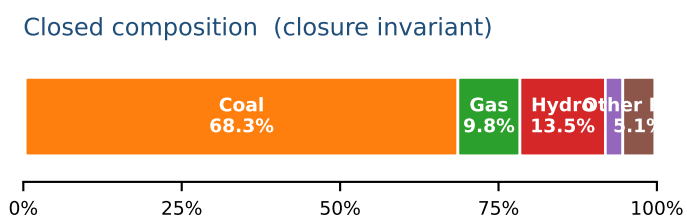




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0030	-1.581
Coal	0.6829	+3.861
Gas	0.0979	+1.919
Hydro	0.1347	+2.238
Nuclear	0.0276	+0.653
Other Fossil	0.0510	+1.267
Solar	0.0000	-6.711
Wind	0.0028	-1.648





closure: Σ = 1.000000 OK

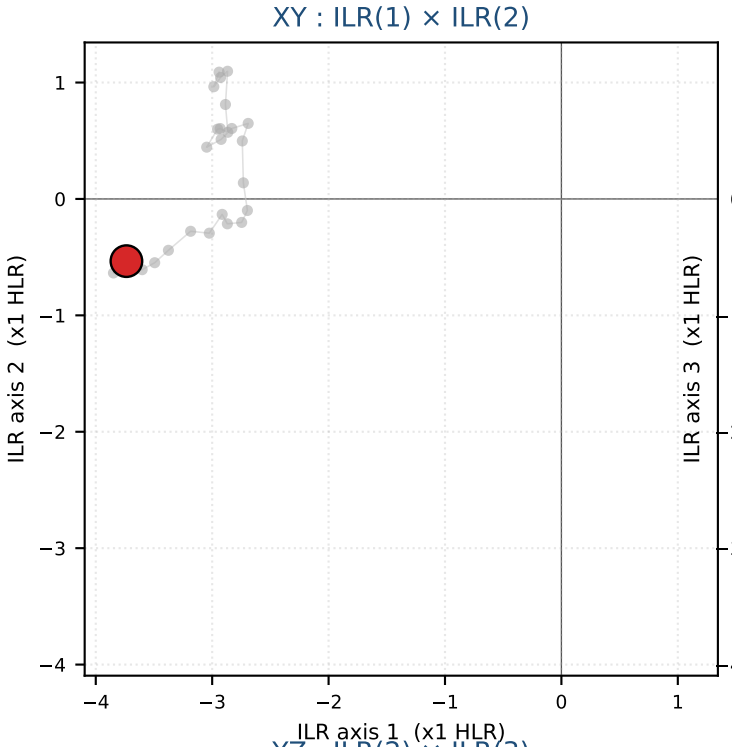
display factor: x1 (raw HLR x 1)

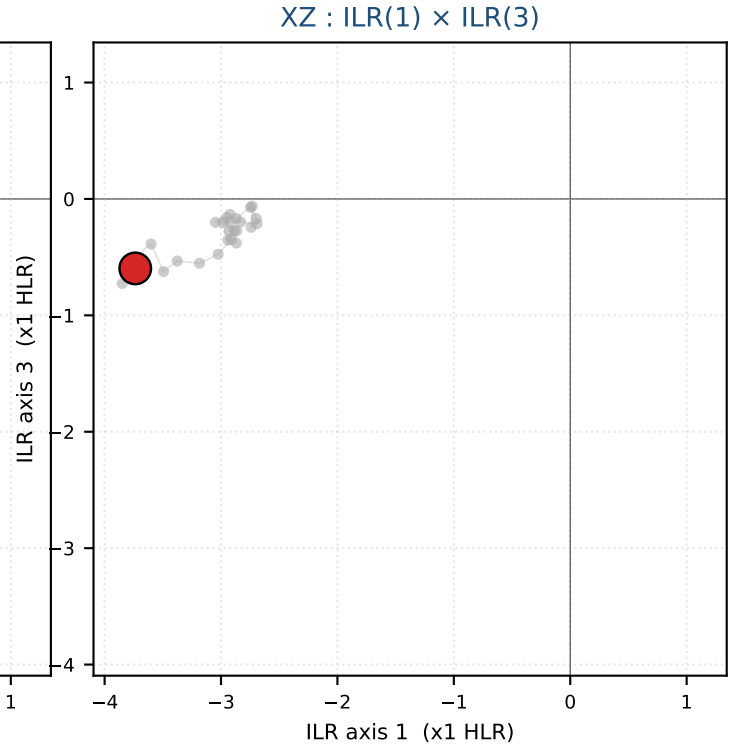
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=1 | 2001 | D=8 T=26 | level 1 first-principles

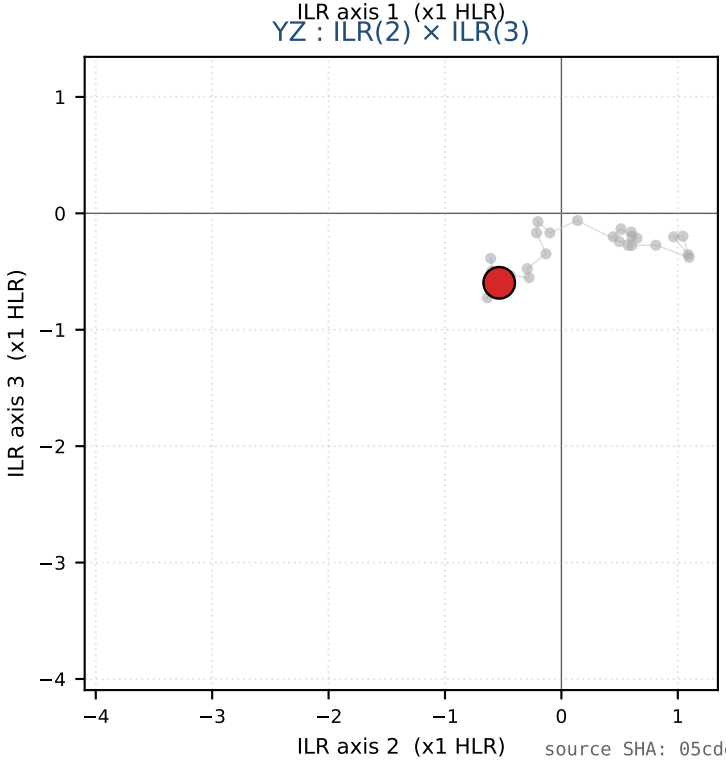
ember_ind | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

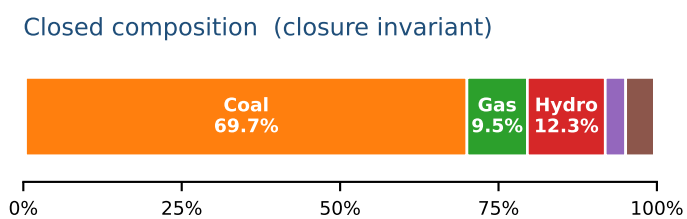




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0035	-1.447
Coal	0.6966	+3.837
Gas	0.0954	+1.849
Hydro	0.1229	+2.102
Nuclear	0.0322	+0.764
Other Fossil	0.0457	+1.113
Solar	0.0000	-6.780
Wind	0.0036	-1.438





closure: Σ = 1.000000 OK

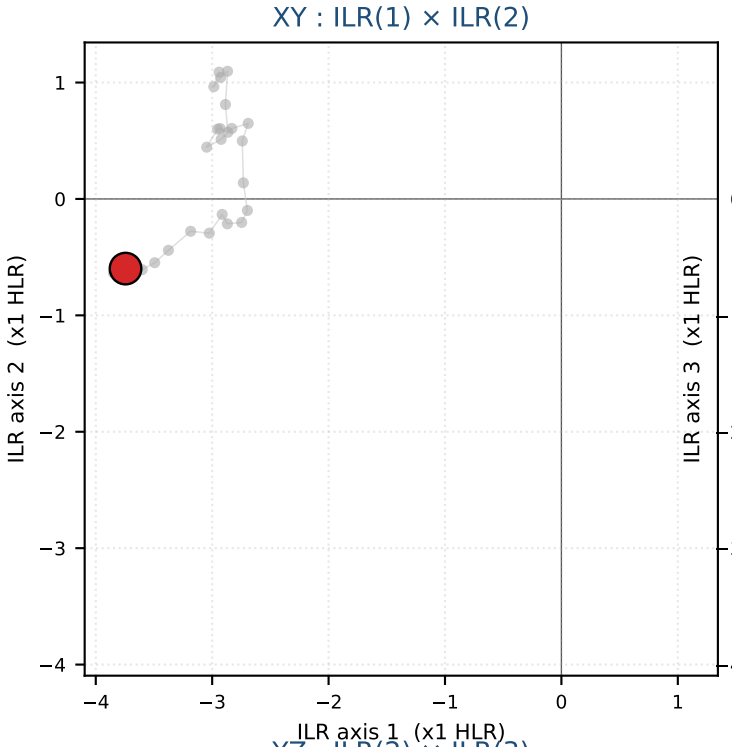
display factor: x1 (raw HLR x 1)

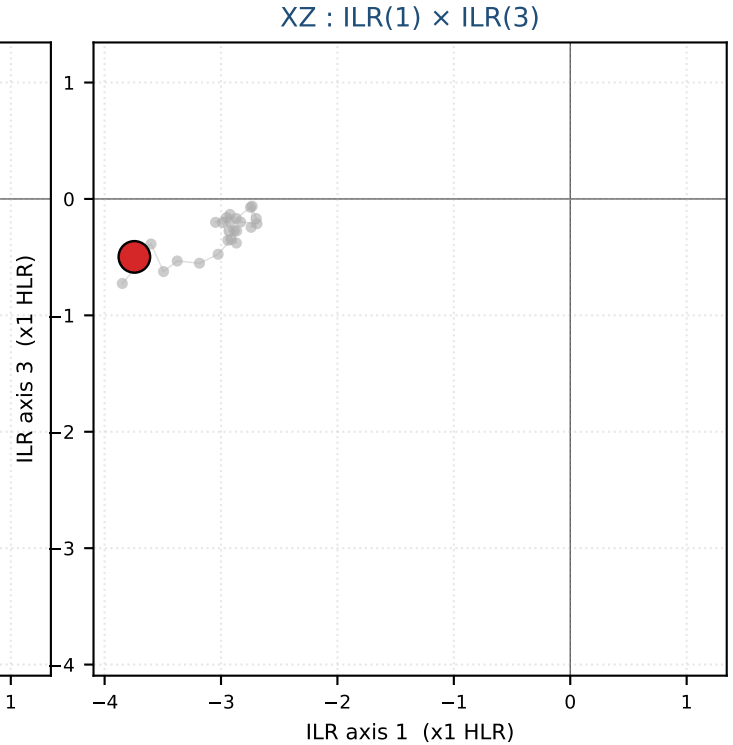
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=2 | 2002 | D=8 T=26 | level 1 first-principles

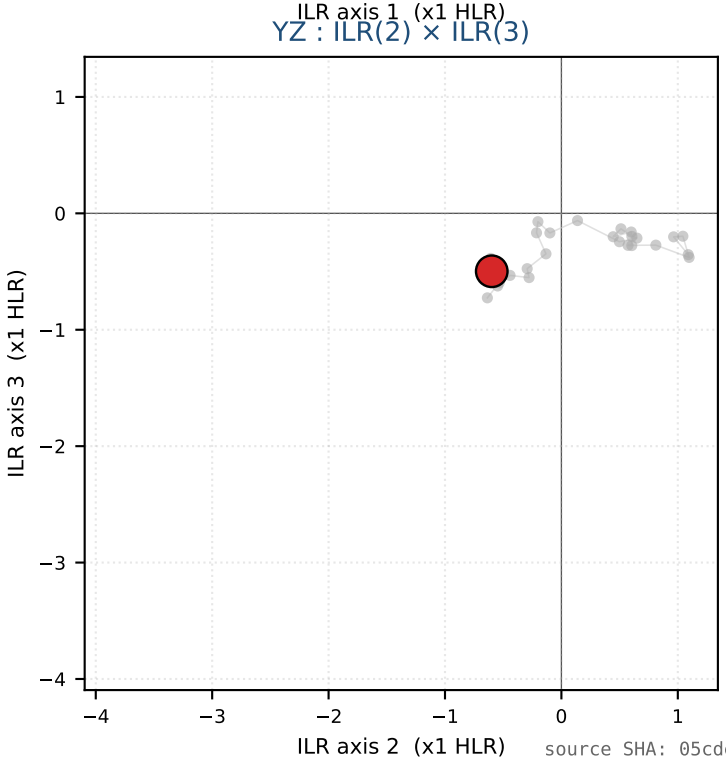
ember_ind | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

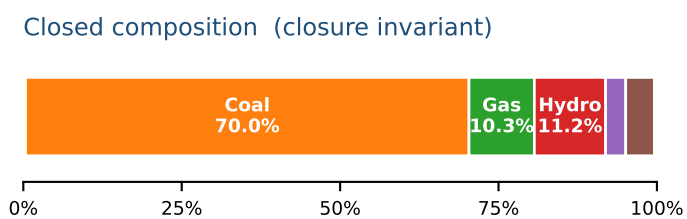




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0035	-1.440
Coal	0.6999	+3.855
Gas	0.1031	+1.940
Hydro	0.1124	+2.026
Nuclear	0.0317	+0.761
Other Fossil	0.0458	+1.128
Solar	0.0000	-6.806
Wind	0.0034	-1.464





closure: Σ = 1.000000 OK

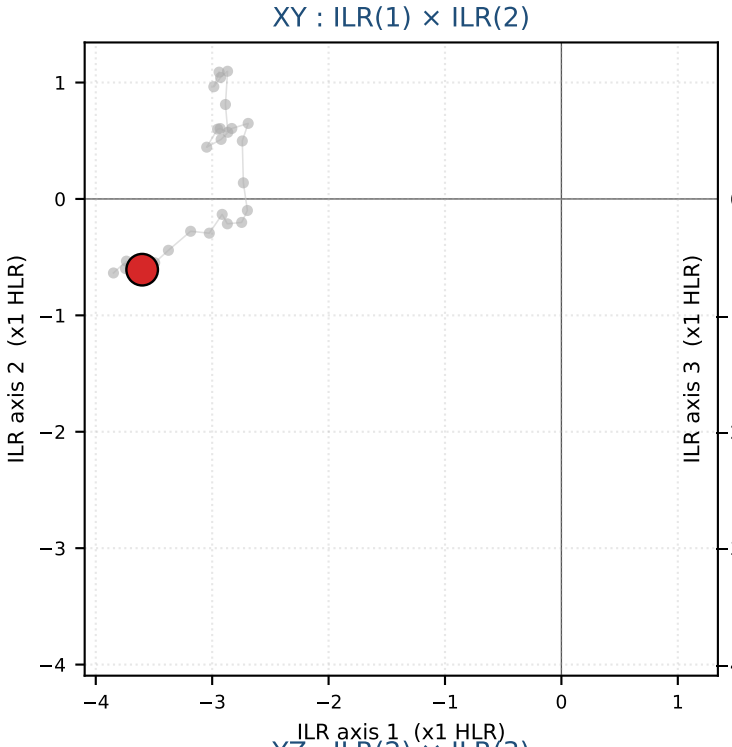
display factor: x1 (raw HLR x 1)

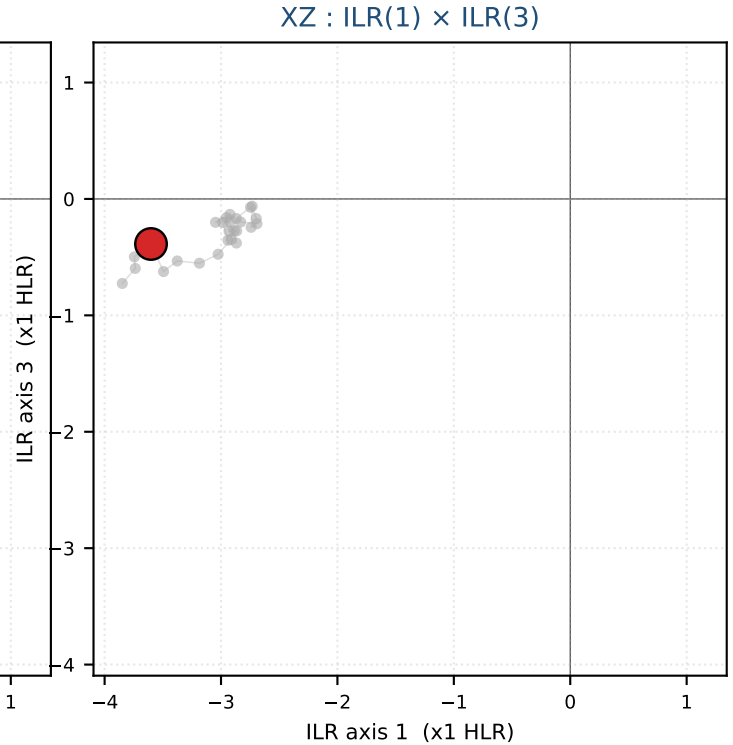
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=3 | 2003 | D=8 T=26 | level 1 first-principles

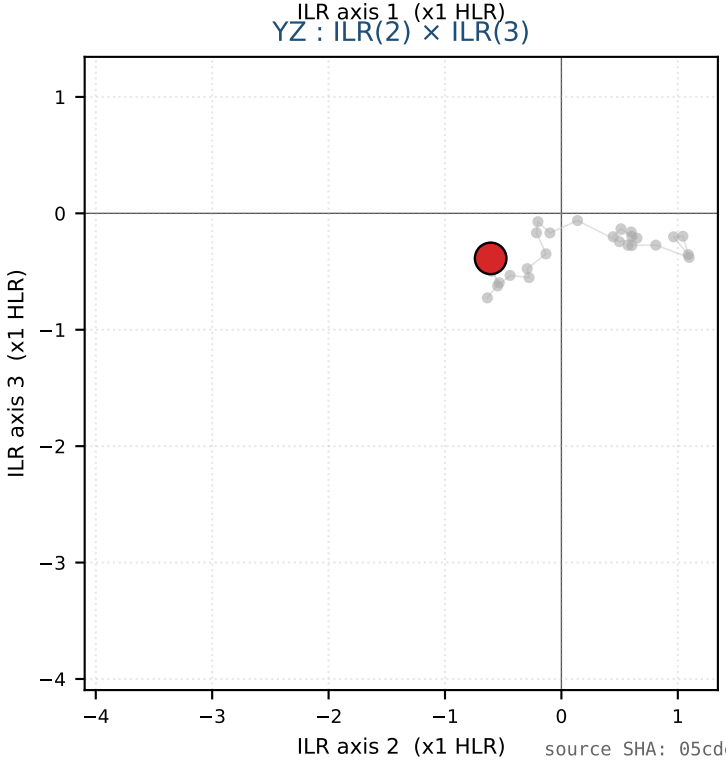
ember_ind | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

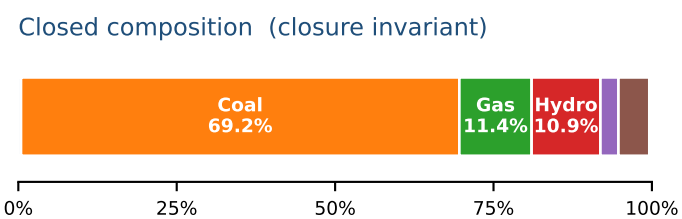




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0042	-1.377
Coal	0.6918	+3.716
Gas	0.1141	+1.914
Hydro	0.1086	+1.864
Nuclear	0.0284	+0.524
Other Fossil	0.0487	+1.063
Solar	0.0000	-6.286
Wind	0.0041	-1.418





closure: Σ = 1.000000 OK

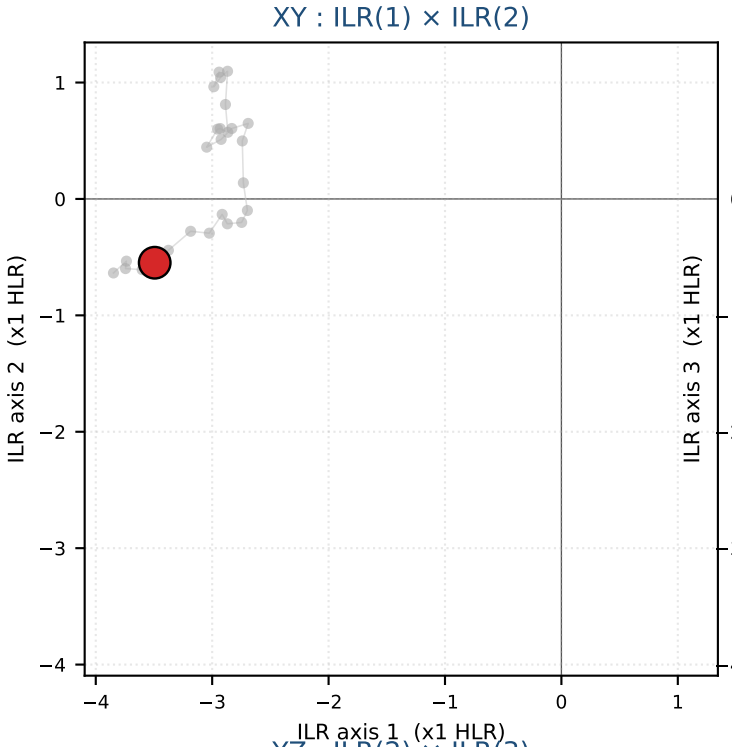
display factor: x1 (raw HLR x 1)

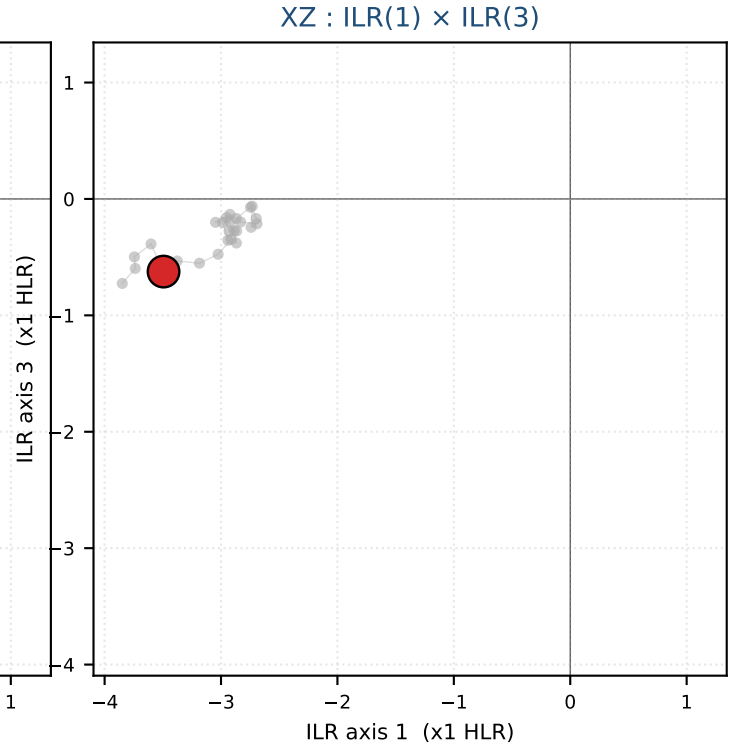
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=4 | 2004 | D=8 T=26 | level 1 first-principles

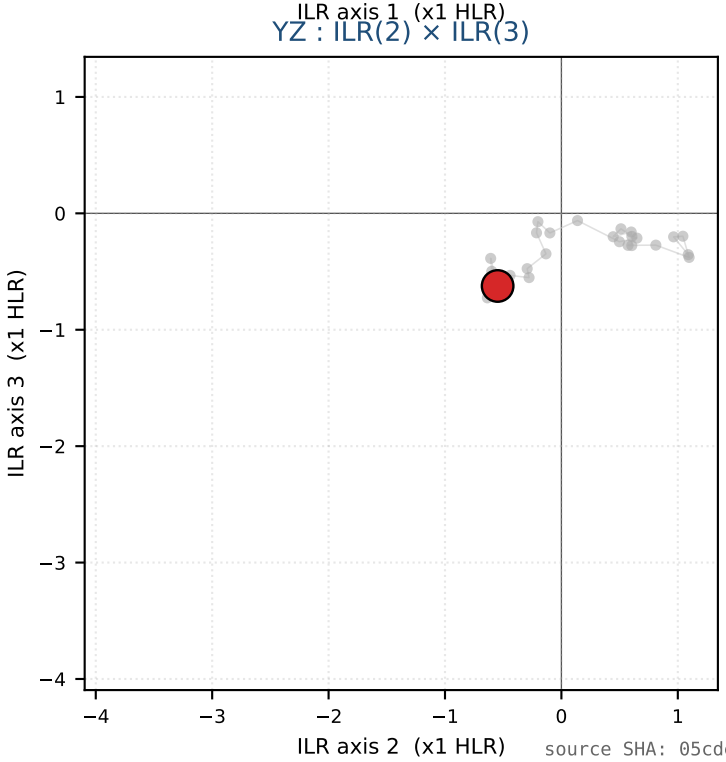
ember_ind | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

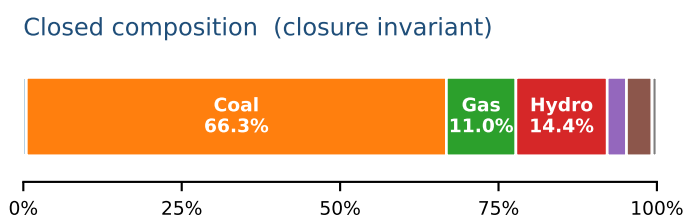




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0047	-1.356
Coal	0.6630	+3.585
Gas	0.1097	+1.785
Hydro	0.1441	+2.058
Nuclear	0.0304	+0.504
Other Fossil	0.0405	+0.789
Solar	0.0000	-6.465
Wind	0.0075	-0.899





closure: Σ = 1.000000 OK

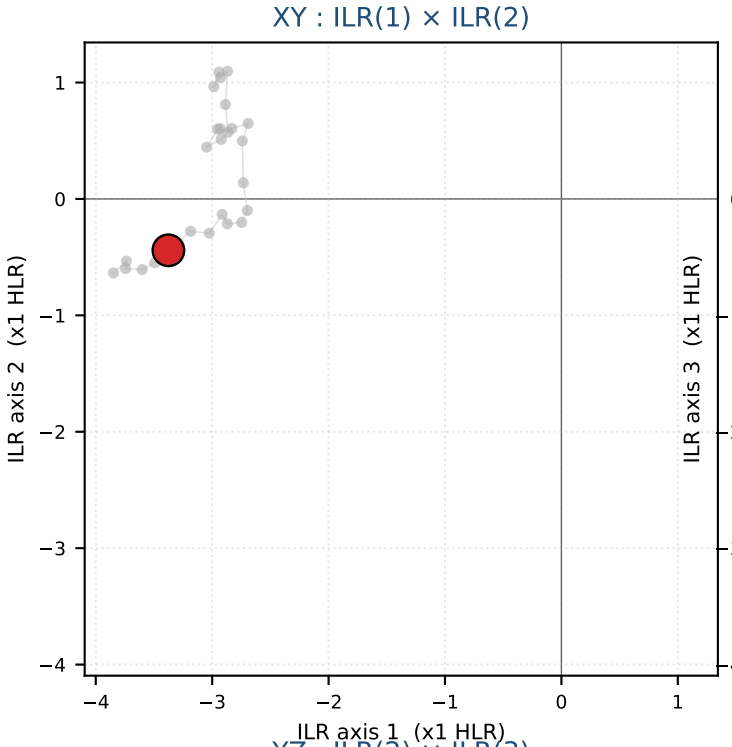
display factor: x1 (raw HLR x 1)

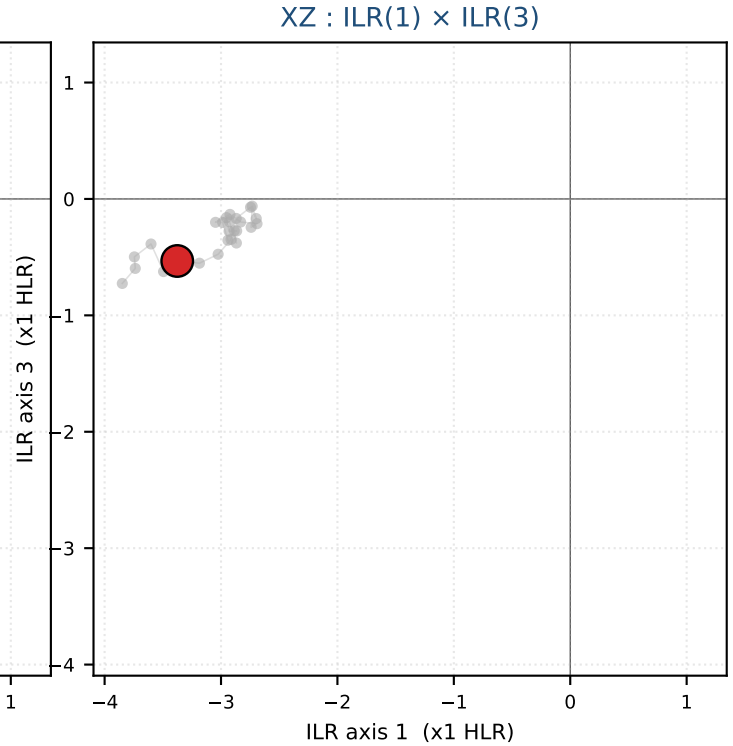
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=5 | 2005 | D=8 T=26 | level 1 first-principles

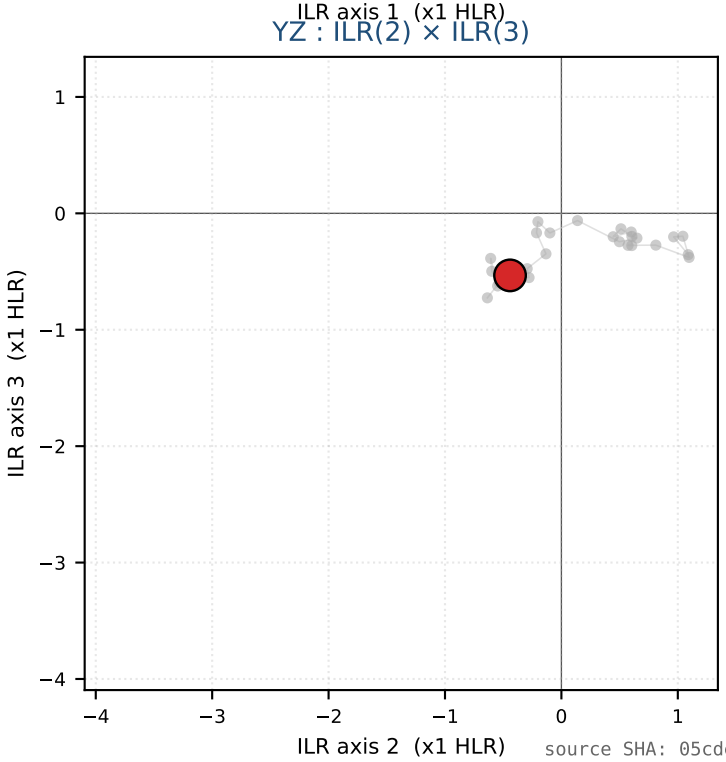
ember_ind | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

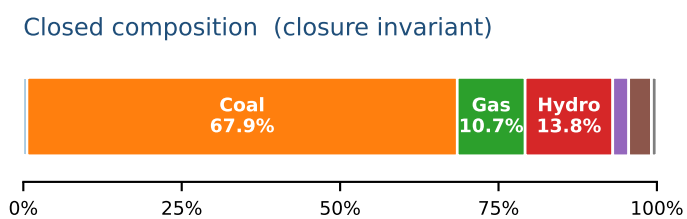




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0057	-1.161
Coal	0.6792	+3.614
Gas	0.1071	+1.767
Hydro	0.1383	+2.022
Nuclear	0.0252	+0.318
Other Fossil	0.0360	+0.677
Solar	0.0000	-6.469
Wind	0.0085	-0.767





closure: Σ = 1.000000 OK

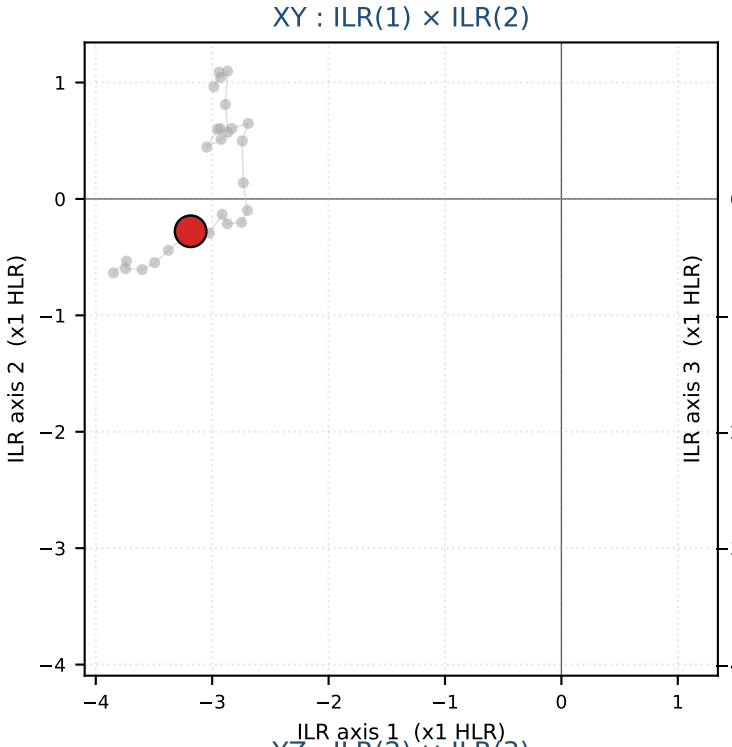
display factor: x1 (raw HLR x 1)

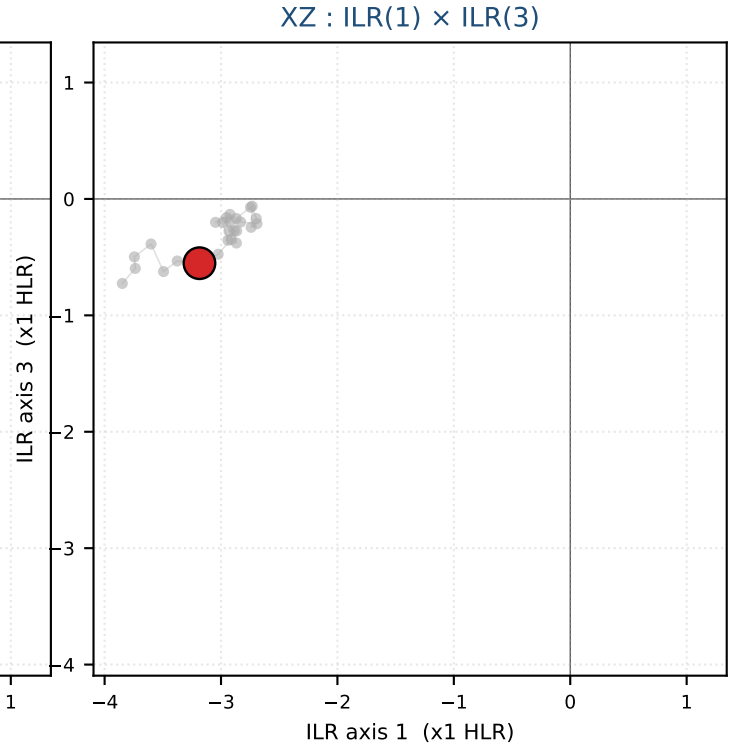
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=6 | 2006 | D=8 T=26 | level 1 first-principles

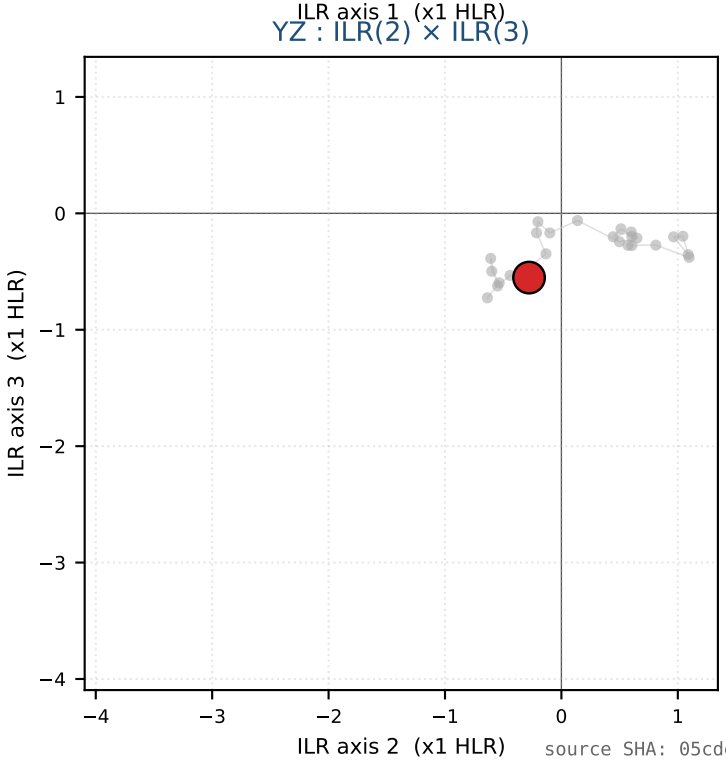
ember_ind | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

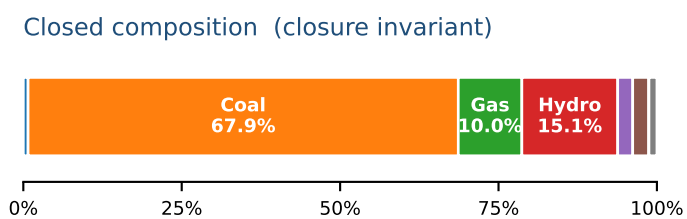




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0075	-0.833
Coal	0.6790	+3.672
Gas	0.1003	+1.760
Hydro	0.1513	+2.170
Nuclear	0.0237	+0.316
Other Fossil	0.0256	+0.395
Solar	0.0000	-7.159
Wind	0.0125	-0.320





closure: Σ = 1.000000 OK

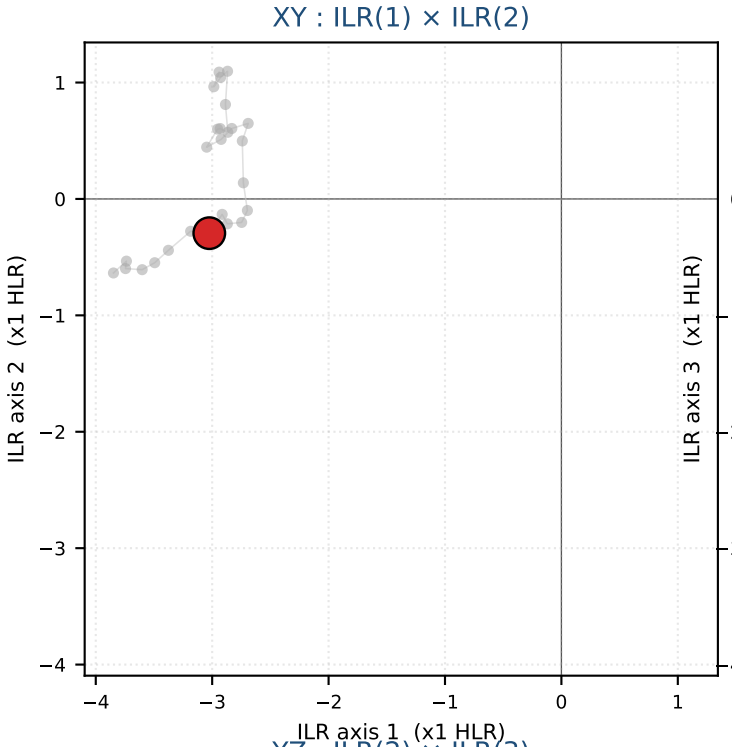
display factor: x1 (raw HLR x 1)

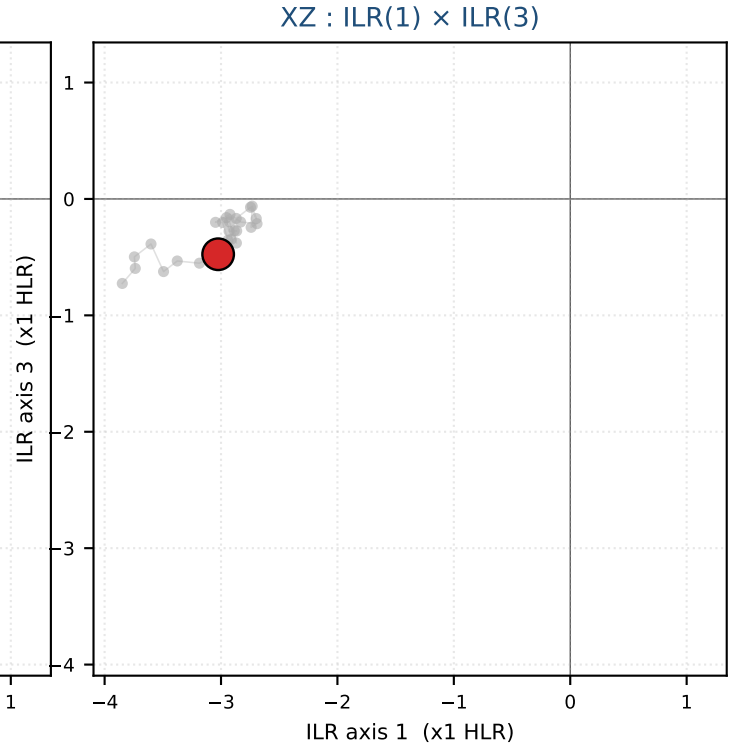
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=7 | 2007 | D=8 T=26 | level 1 first-principles

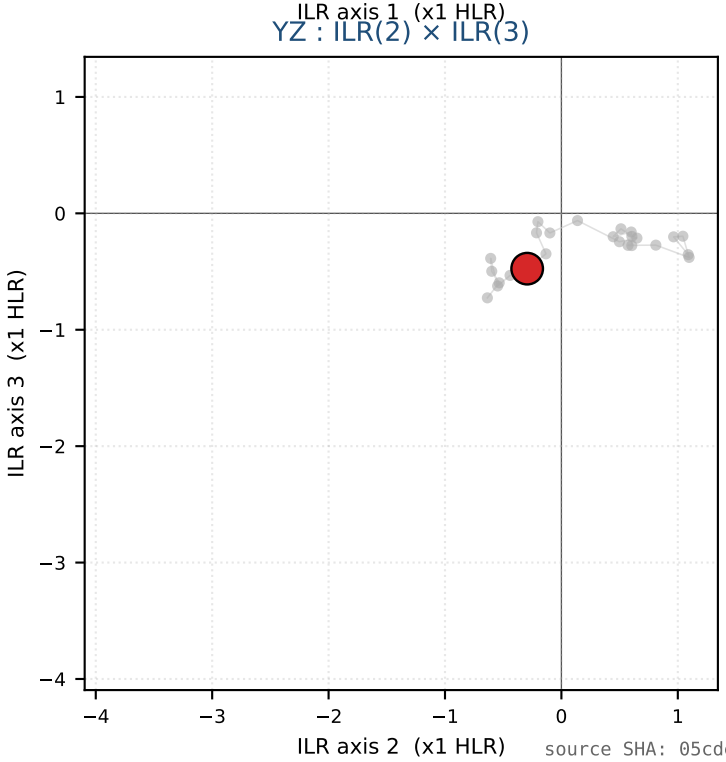
ember_ind | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

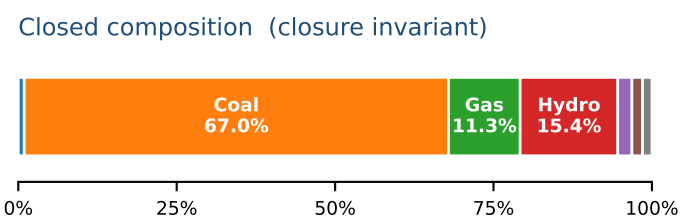




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0093	-0.837
Coal	0.6697	+3.441
Gas	0.1131	+1.662
Hydro	0.1540	+1.971
Nuclear	0.0224	+0.042
Other Fossil	0.0168	-0.246
Solar	0.0001	-5.652
Wind	0.0147	-0.381





closure: Σ = 1.000000 OK

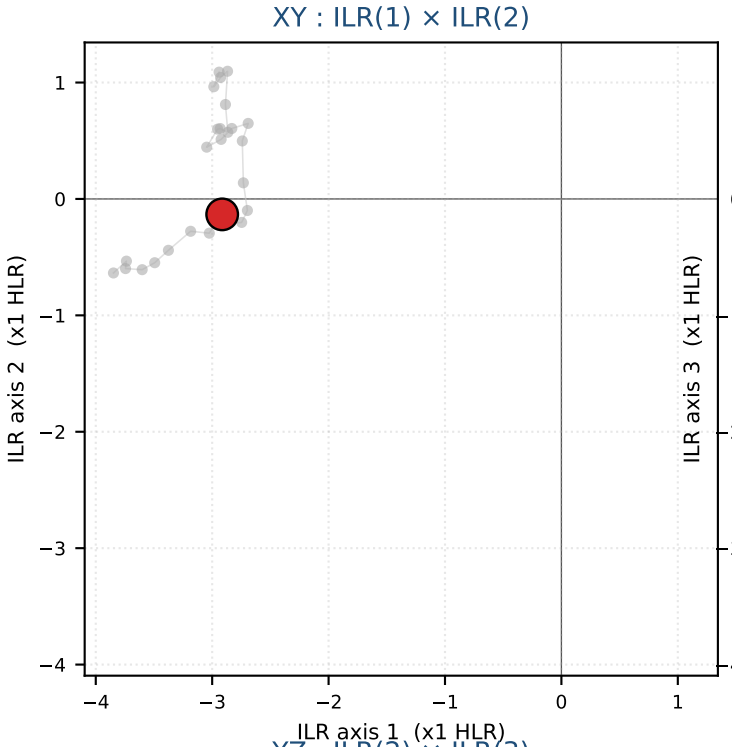
display factor: x1 (raw HLR x 1)

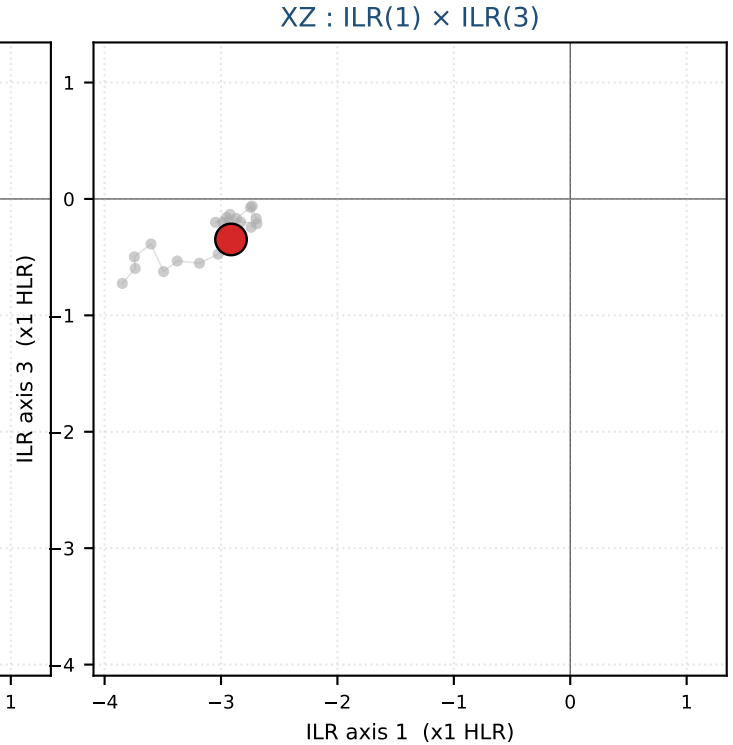
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=8 | 2008 | D=8 T=26 | level 1 first-principles

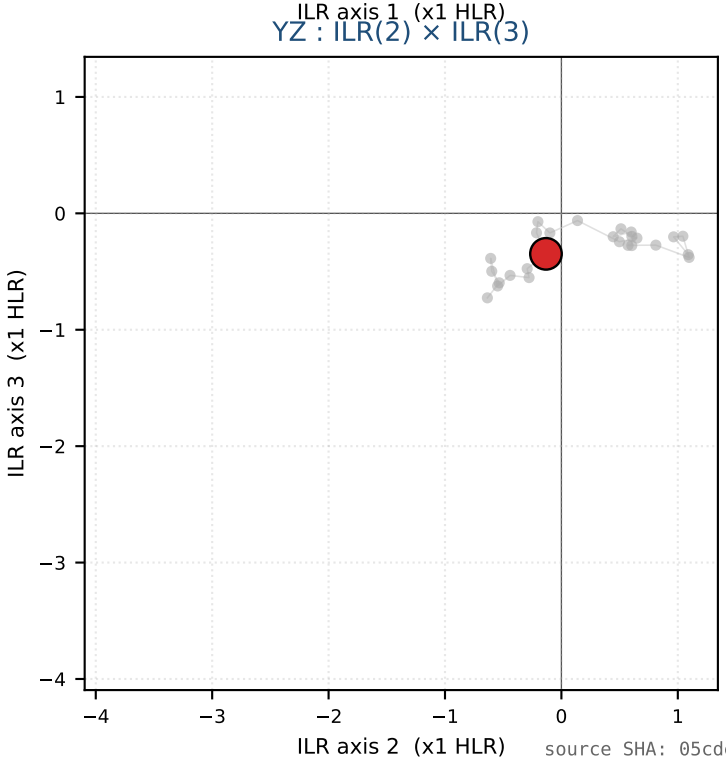
ember_ind | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

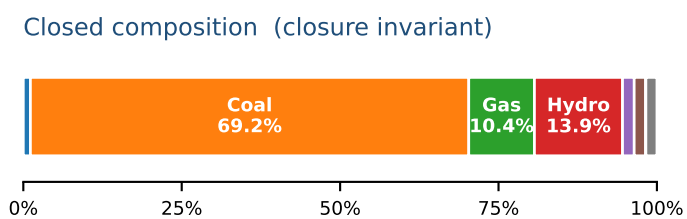




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0112	-0.655
Coal	0.6918	+3.466
Gas	0.1037	+1.568
Hydro	0.1390	+1.861
Nuclear	0.0184	-0.162
Other Fossil	0.0184	-0.159
Solar	0.0001	-5.699
Wind	0.0174	-0.219





closure: Σ = 1.000000 OK

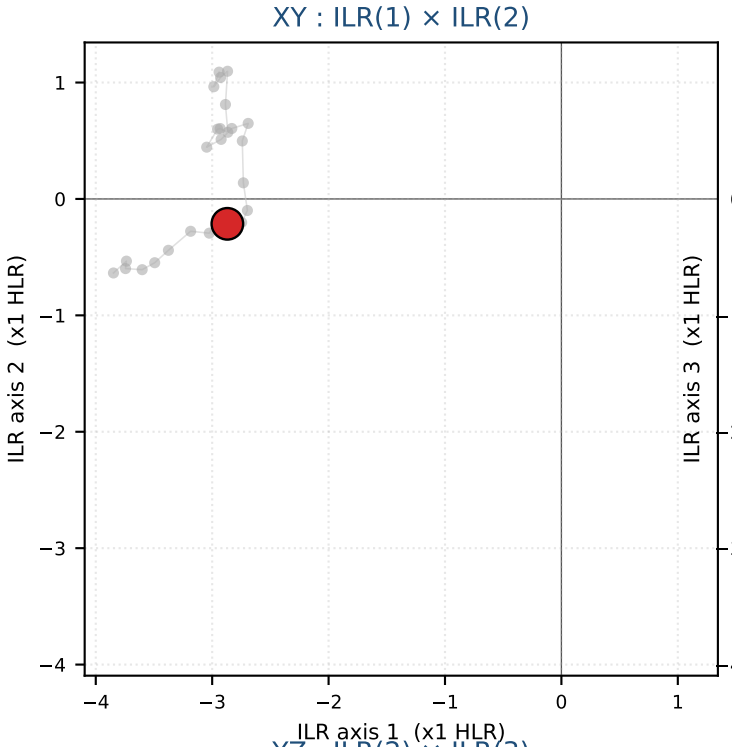
display factor: x1 (raw HLR x 1)

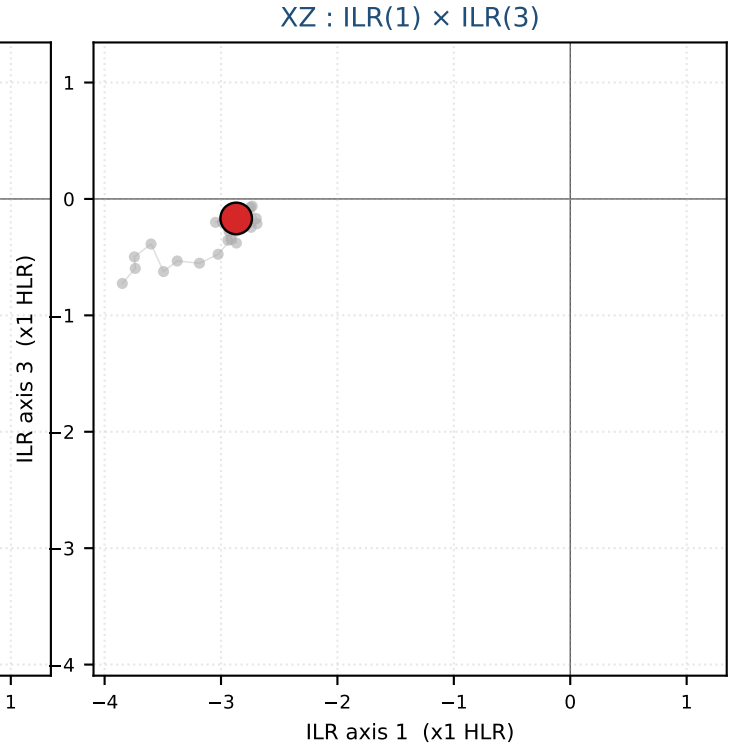
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=9 | 2009 | D=8 T=26 | level 1 first-principles

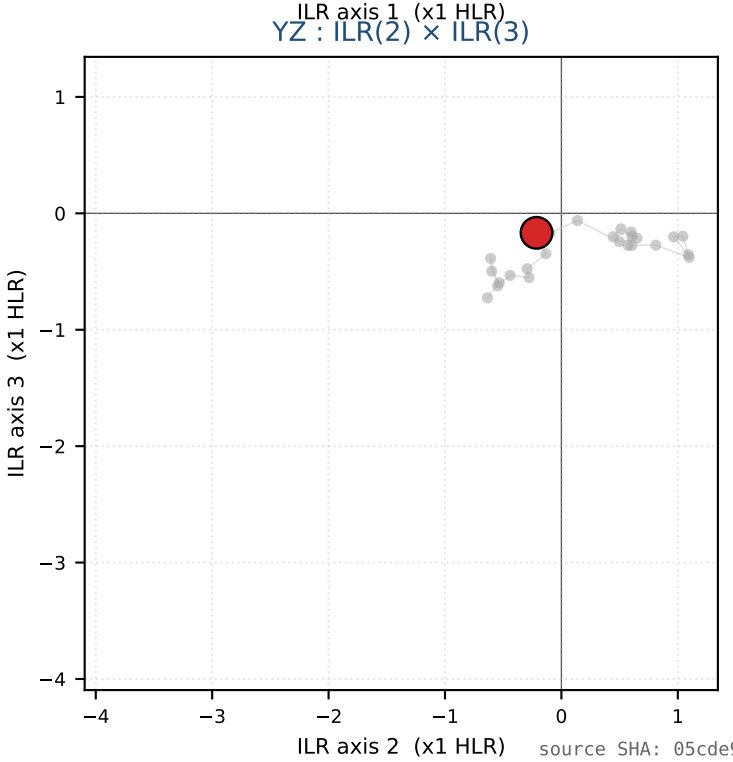
ember_ind | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

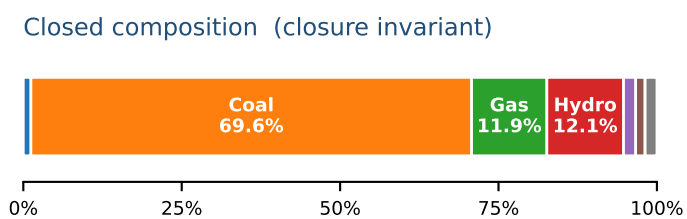




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0120	-0.611
Coal	0.6955	+3.447
Gas	0.1188	+1.680
Hydro	0.1210	+1.699
Nuclear	0.0191	-0.145
Other Fossil	0.0149	-0.397
Solar	0.0001	-5.494
Wind	0.0185	-0.179





closure: Σ = 1.000000 OK

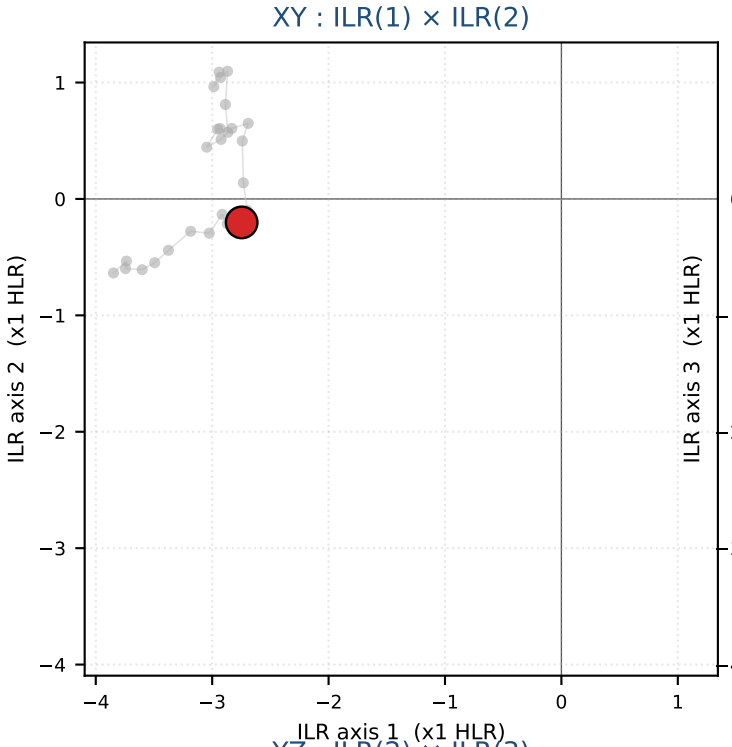
display factor: x1 (raw HLR x 1)

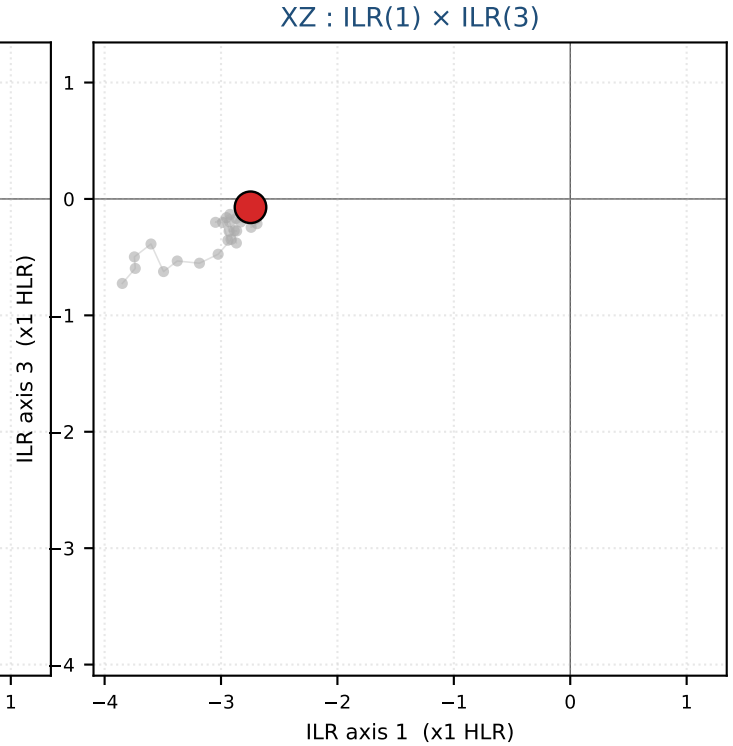
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=10 | 2010 | D=8 T=26 | level 1 first-principles

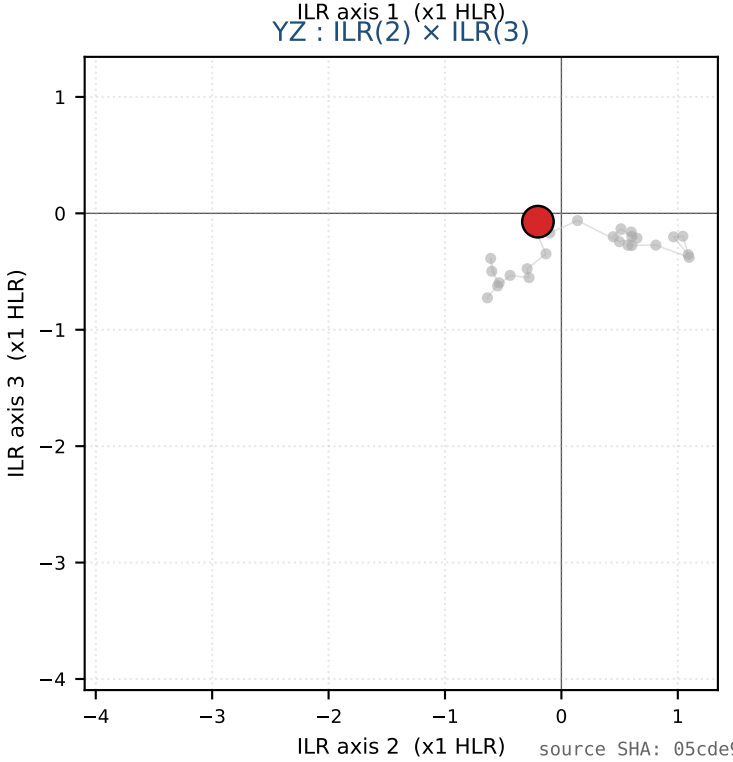
ember_ind | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4



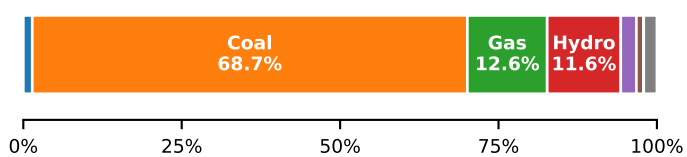


Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0141	-0.516
Coal	0.6866	+3.368
Gas	0.1260	+1.673
Hydro	0.1161	+1.591
Nuclear	0.0246	+0.041
Other Fossil	0.0115	-0.720
Solar	0.0001	-5.305
Wind	0.0208	-0.130



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

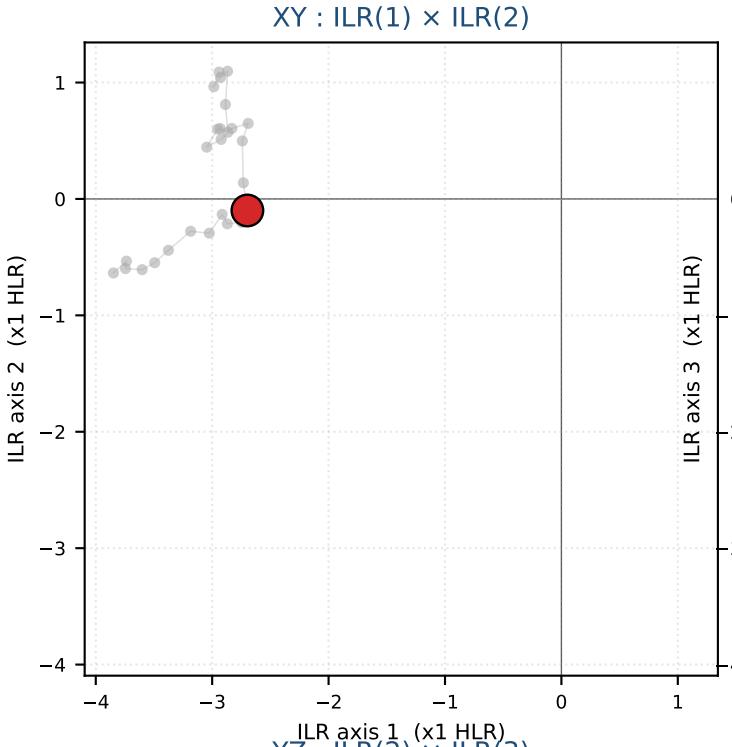
display factor: x1 (raw HLR x 1)

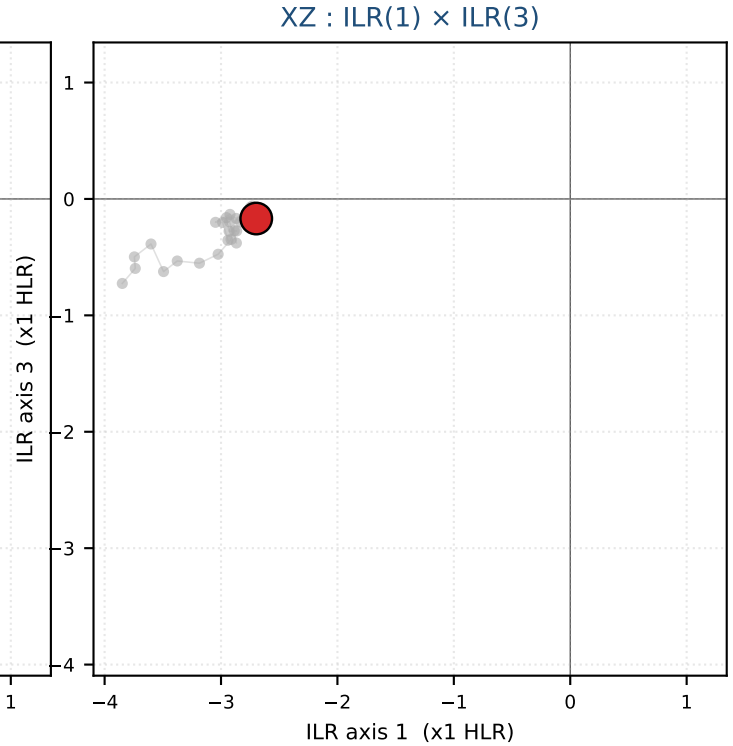
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=11 | 2011 | D=8 T=26 | level 1 first-principles

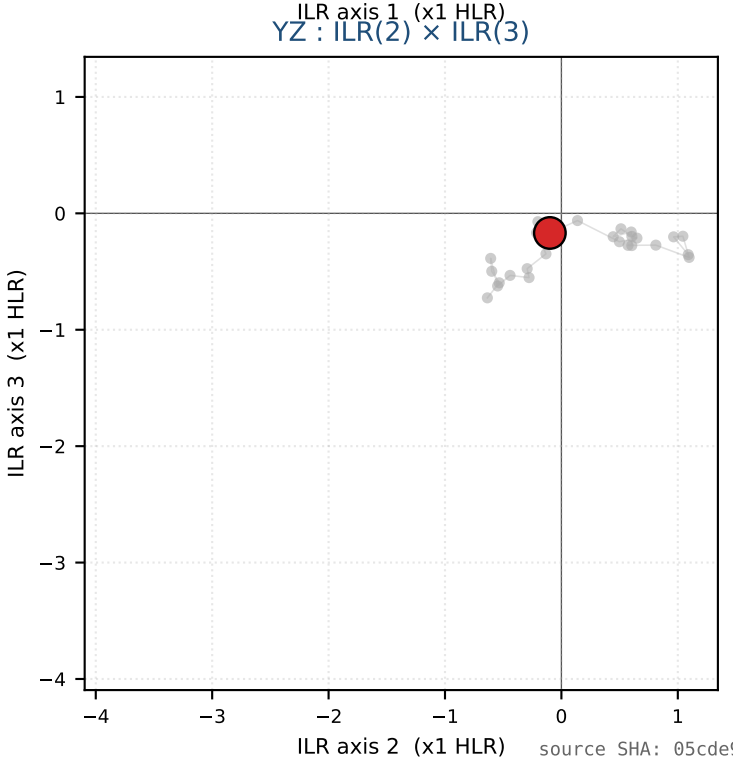
ember_ind | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

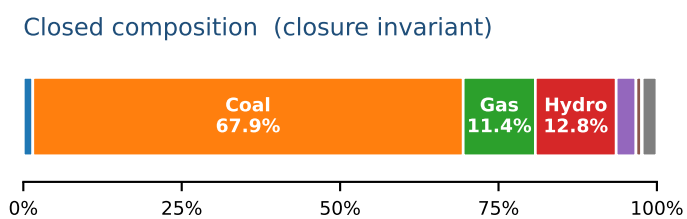




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0150	-0.716
Coal	0.6793	+3.098
Gas	0.1139	+1.312
Hydro	0.1276	+1.426
Nuclear	0.0312	+0.018
Other Fossil	0.0091	-1.219
Solar	0.0008	-3.641
Wind	0.0232	-0.279





closure: Σ = 1.000000 OK

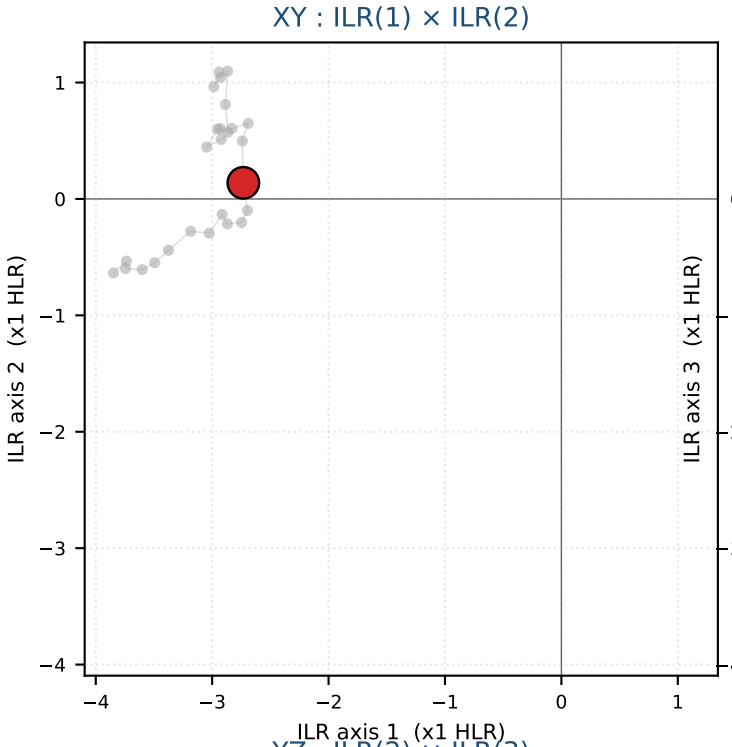
display factor: x1 (raw HLR x 1)

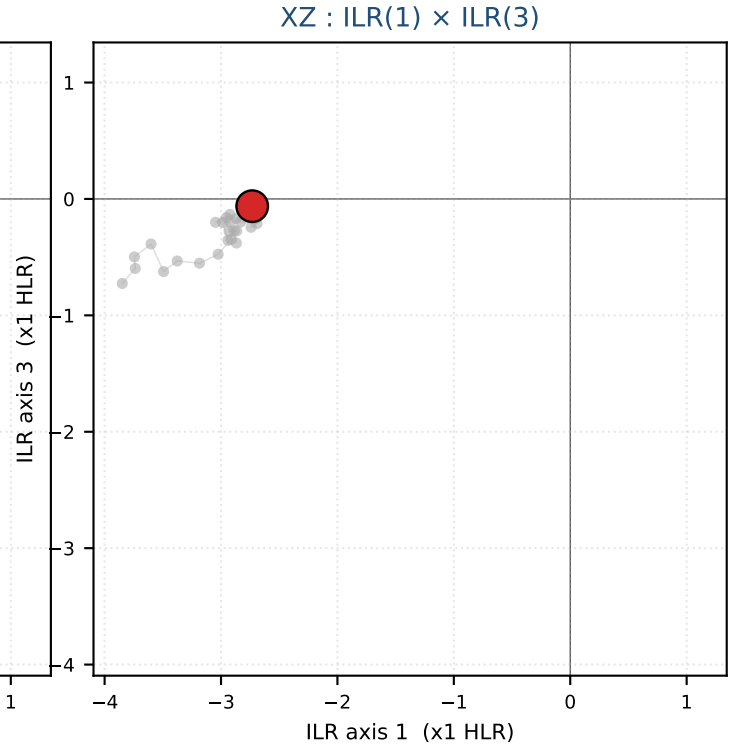
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=12 | 2012 | D=8 T=26 | level 1 first-principles

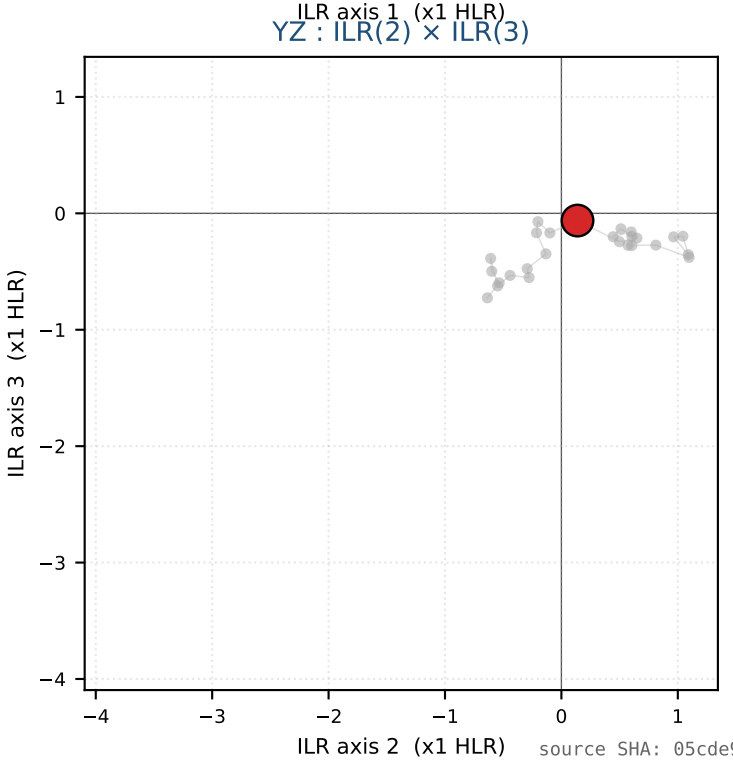
ember_ind | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

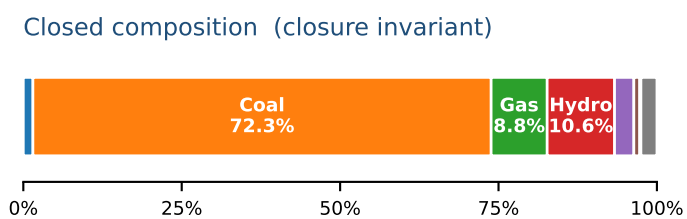




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0152	-0.777
Coal	0.7232	+3.086
Gas	0.0884	+0.985
Hydro	0.1064	+1.170
Nuclear	0.0304	-0.081
Other Fossil	0.0093	-1.272
Solar	0.0019	-2.840
Wind	0.0252	-0.270





closure: Σ = 1.000000 OK

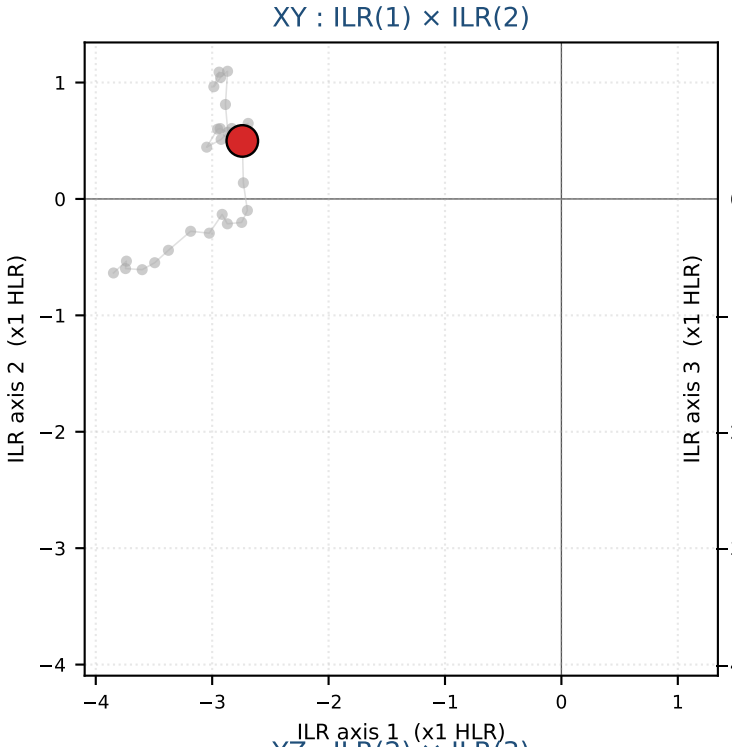
display factor: x1 (raw HLR x 1)

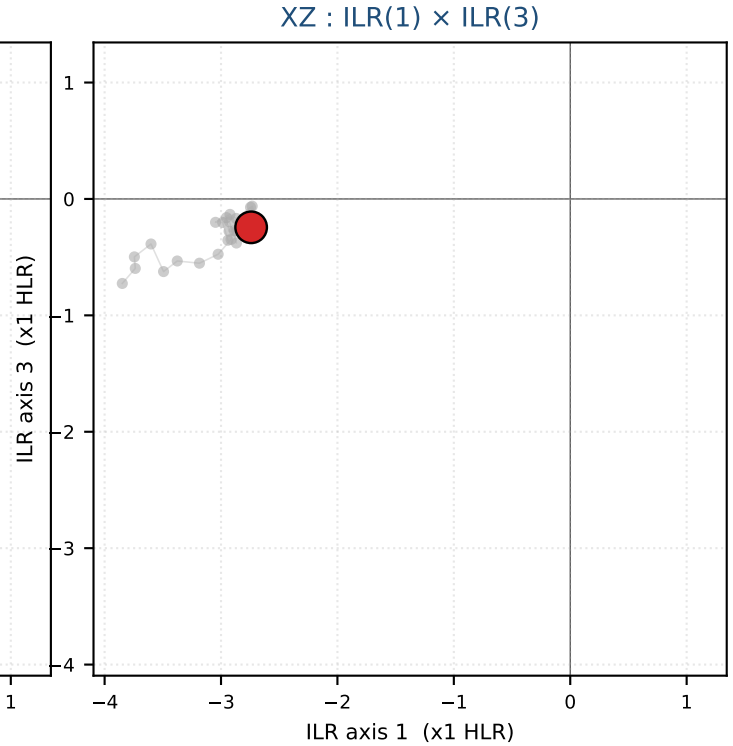
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=13 | 2013 | D=8 T=26 | level 1 first-principles

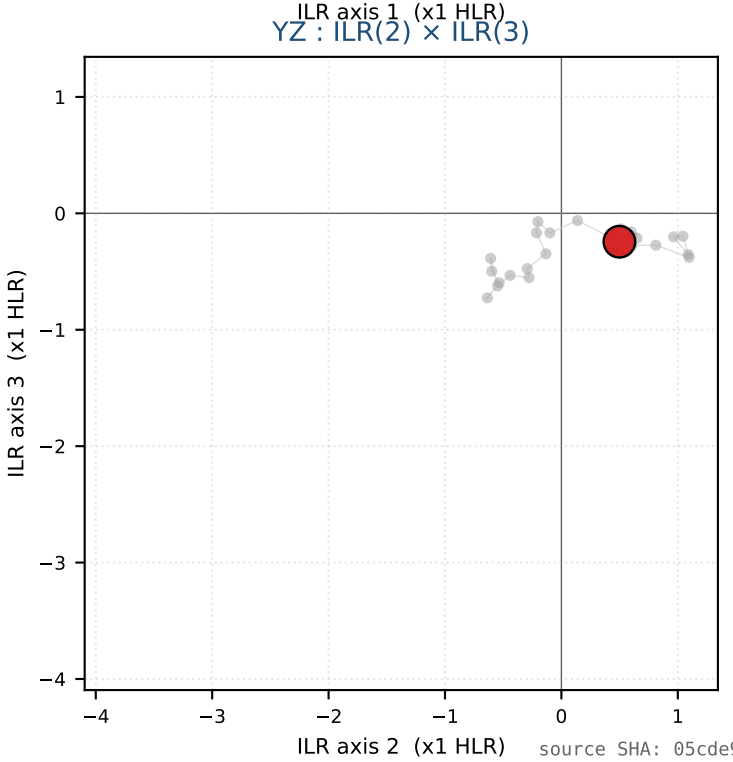
ember_ind | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

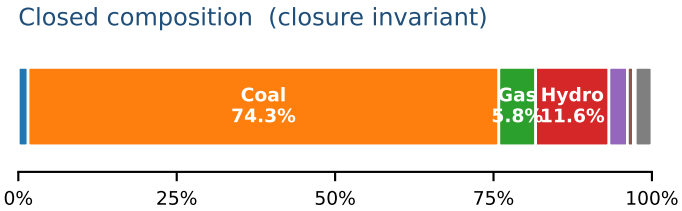




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0154	-0.783
Coal	0.7430	+3.094
Gas	0.0581	+0.545
Hydro	0.1156	+1.234
Nuclear	0.0292	-0.143
Other Fossil	0.0093	-1.283
Solar	0.0030	-2.416
Wind	0.0263	-0.247





closure: Σ = 1.000000 OK

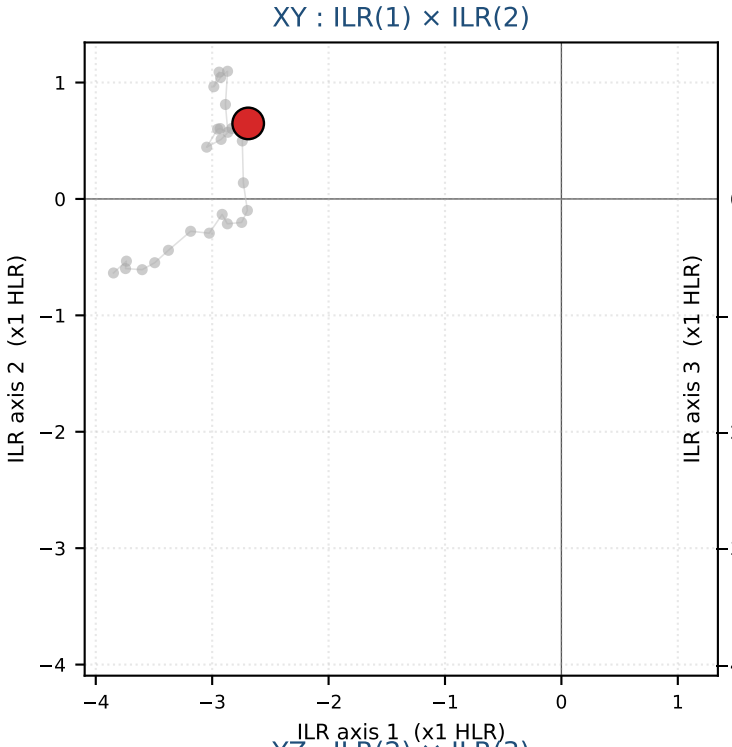
display factor: x1 (raw HLR x 1)

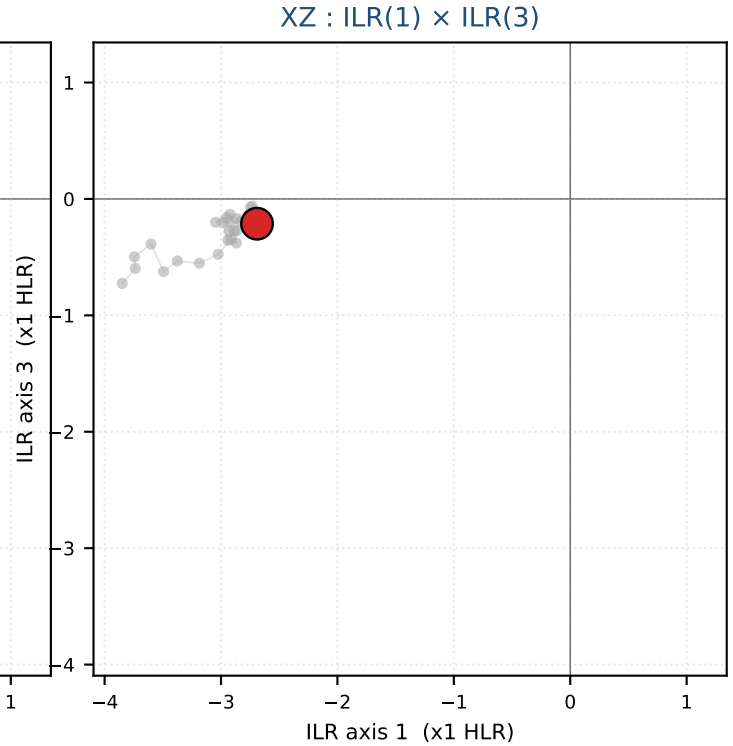
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=14 | 2014 | D=8 T=26 | level 1 first-principles

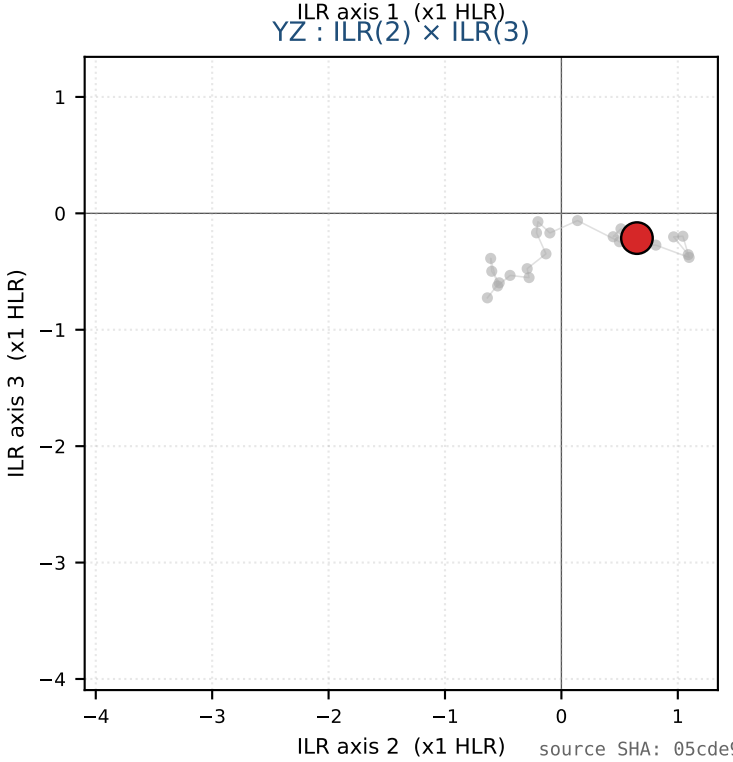
ember_ind | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

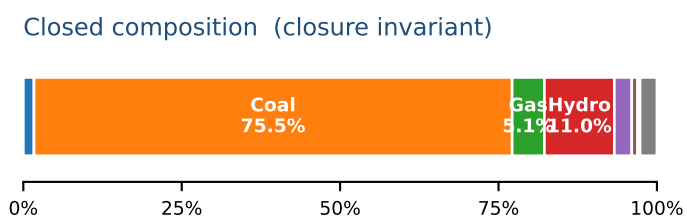




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0168	-0.711
Coal	0.7547	+3.094
Gas	0.0509	+0.397
Hydro	0.1104	+1.173
Nuclear	0.0276	-0.215
Other Fossil	0.0092	-1.315
Solar	0.0039	-2.171
Wind	0.0266	-0.252





closure: Σ = 1.000000 OK

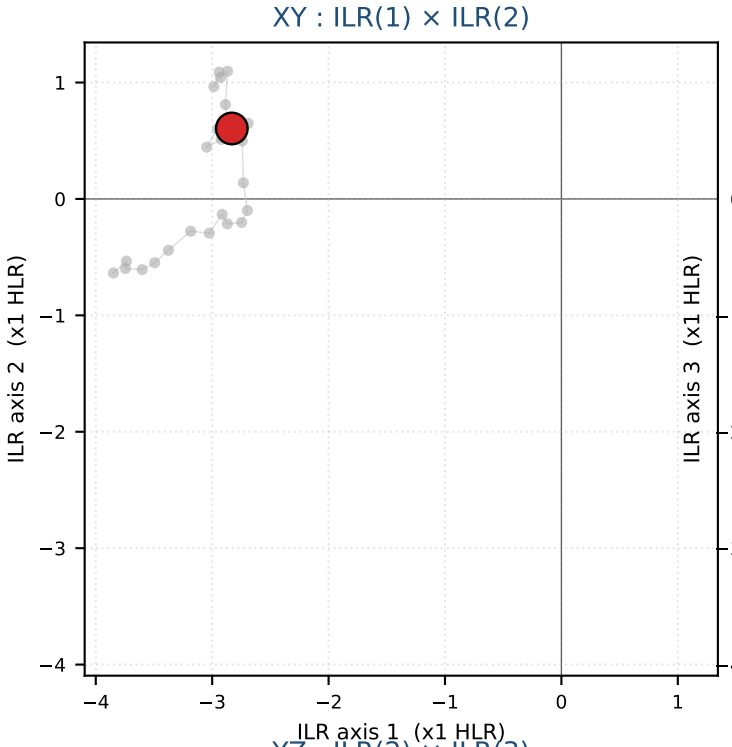
display factor: x1 (raw HLR x 1)

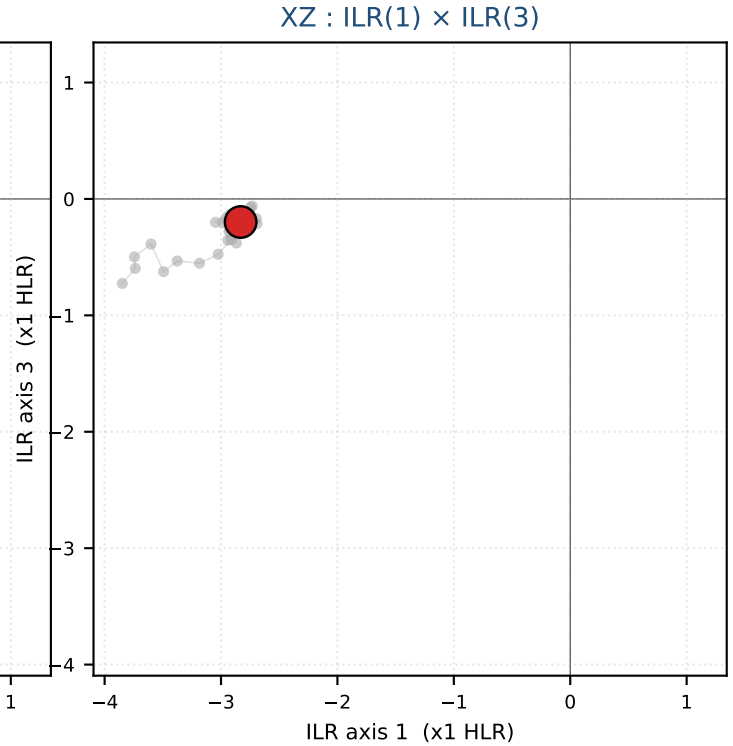
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=15 | 2015 | D=8 T=26 | level 1 first-principles

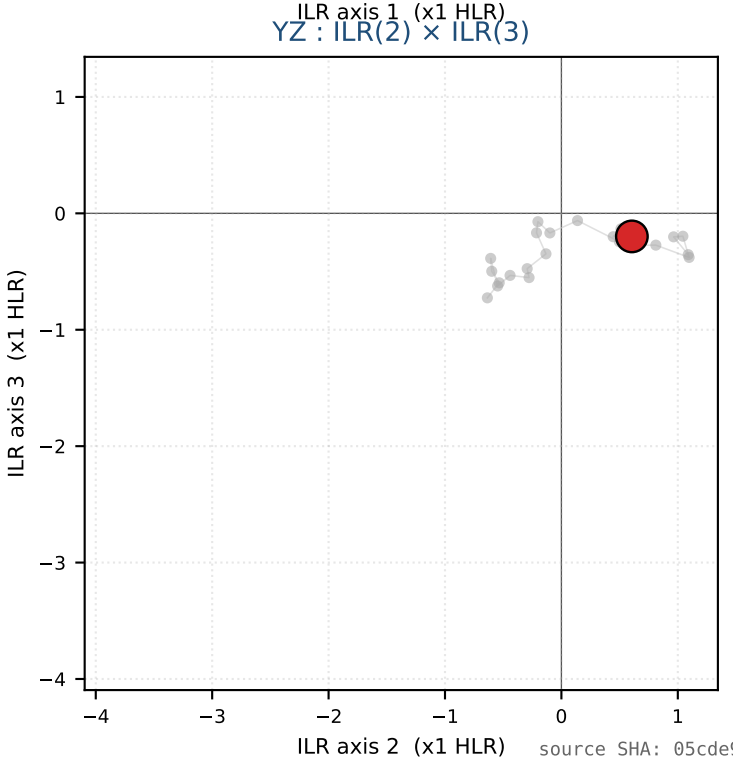
ember_ind | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

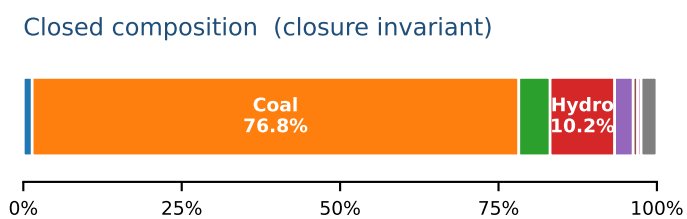




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0140	-0.866
Coal	0.7680	+3.138
Gas	0.0494	+0.395
Hydro	0.1018	+1.118
Nuclear	0.0292	-0.130
Other Fossil	0.0071	-1.550
Solar	0.0054	-1.817
Wind	0.0250	-0.287





closure: Σ = 1.000000 OK

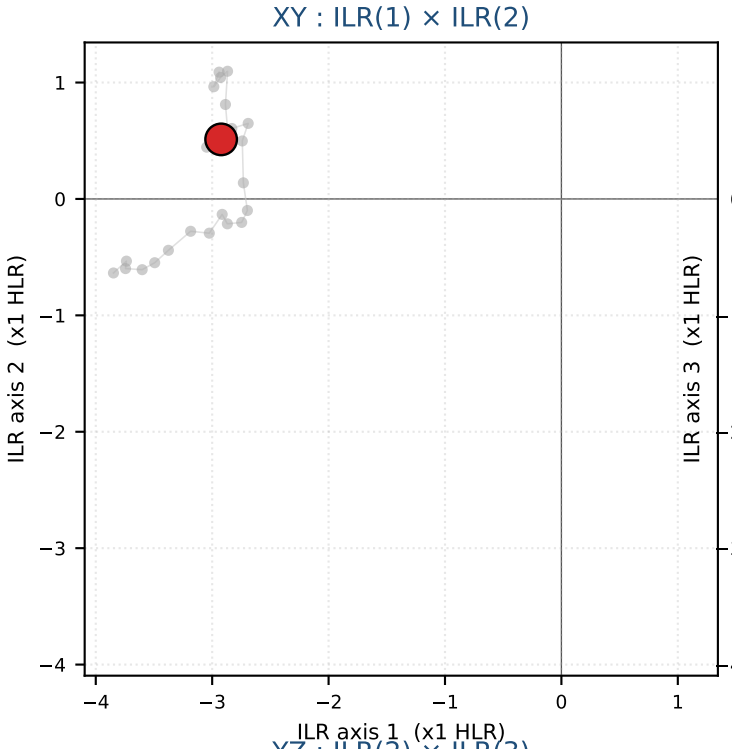
display factor: x1 (raw HLR x 1)

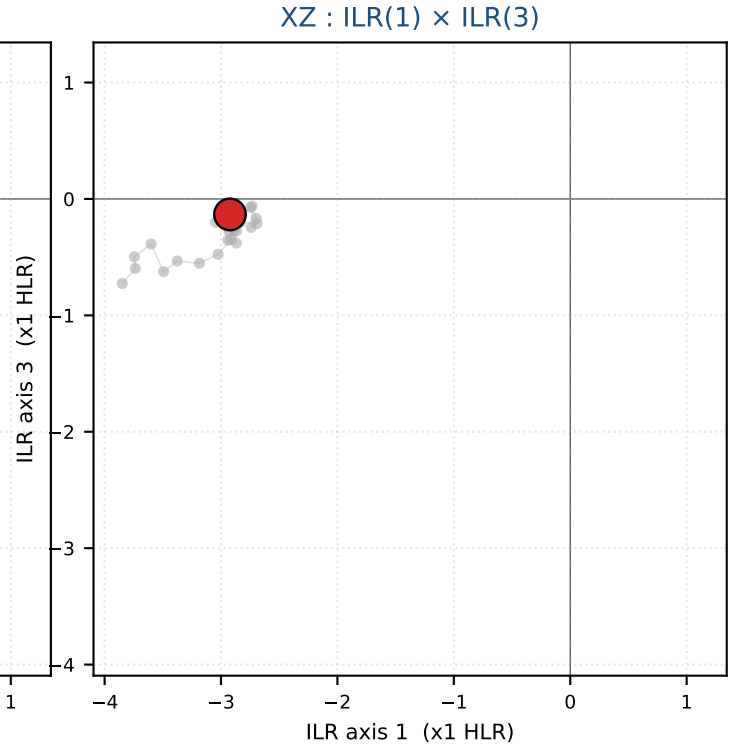
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=16 | 2016 | D=8 T=26 | level 1 first-principles

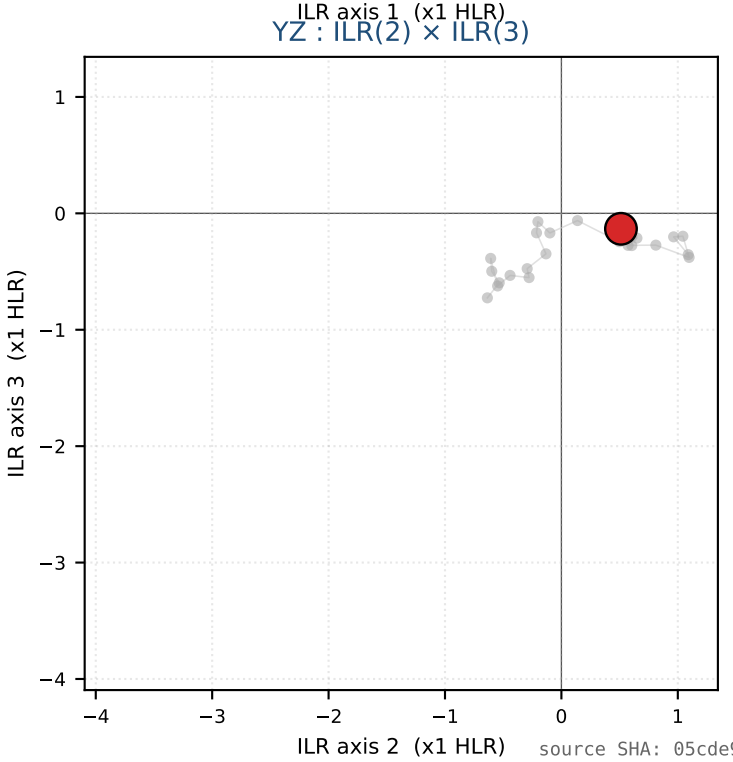
ember_ind | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

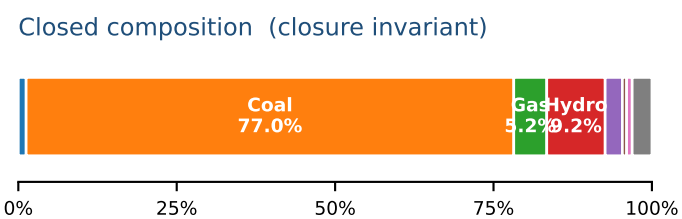




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0123	-1.044
Coal	0.7696	+3.089
Gas	0.0521	+0.396
Hydro	0.0922	+0.967
Nuclear	0.0272	-0.254
Other Fossil	0.0067	-1.648
Solar	0.0087	-1.388
Wind	0.0311	-0.118





closure: Σ = 1.000000 OK

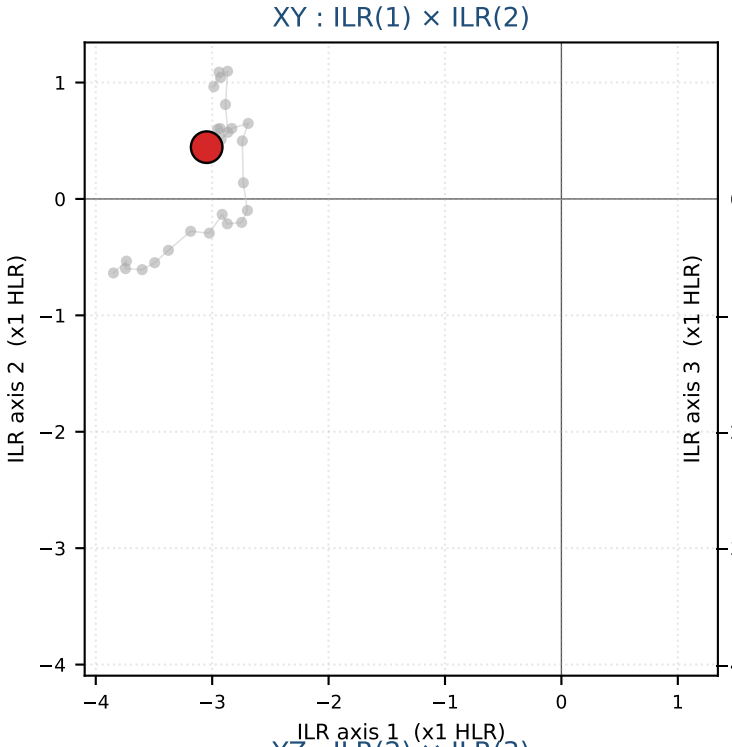
display factor: x1 (raw HLR x 1)

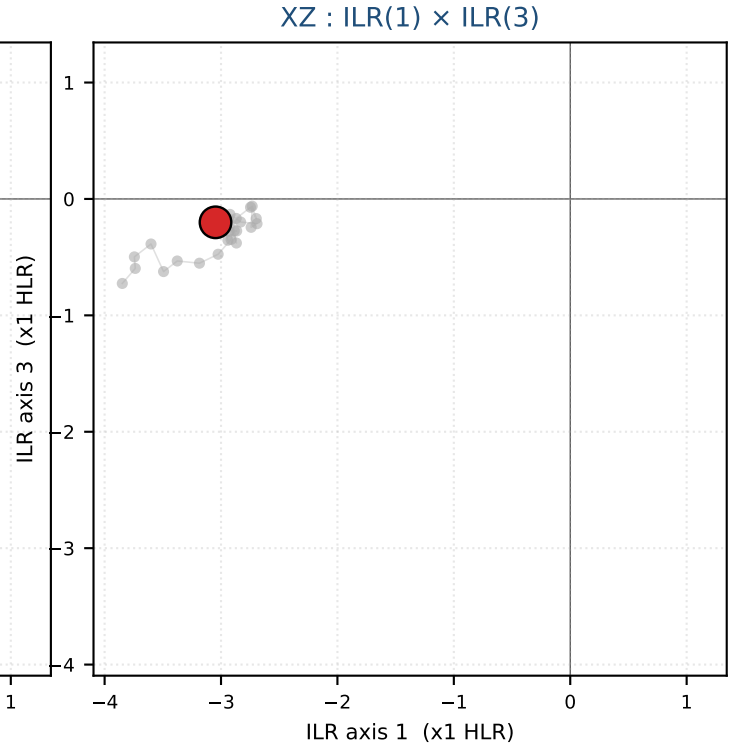
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=17 | 2017 | D=8 T=26 | level 1 first-principles

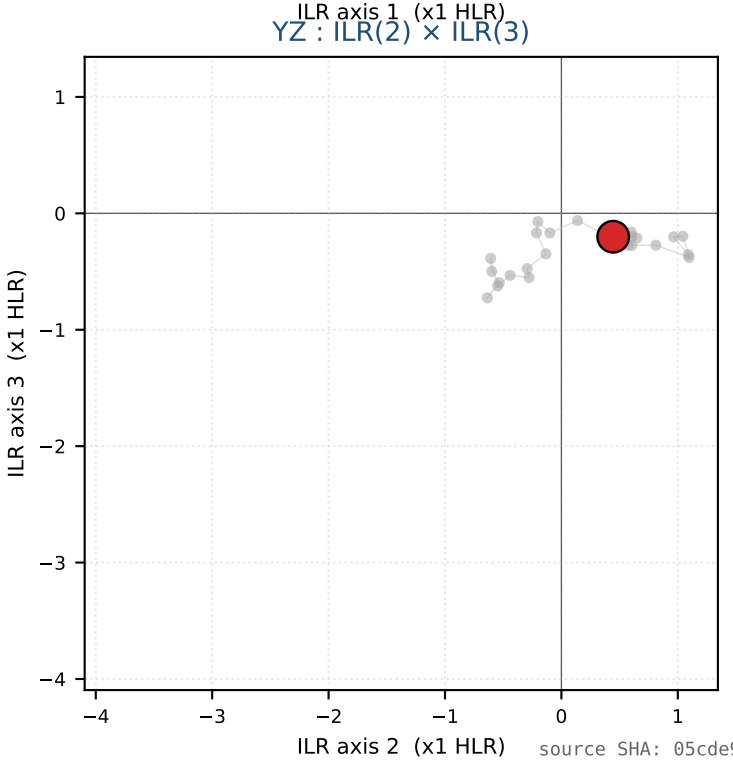
ember_ind | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

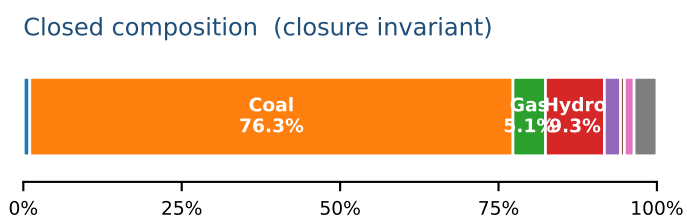




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0103	-1.270
Coal	0.7626	+3.039
Gas	0.0513	+0.340
Hydro	0.0930	+0.935
Nuclear	0.0256	-0.356
Other Fossil	0.0060	-1.804
Solar	0.0153	-0.870
Wind	0.0360	-0.014





closure: Σ = 1.000000 OK

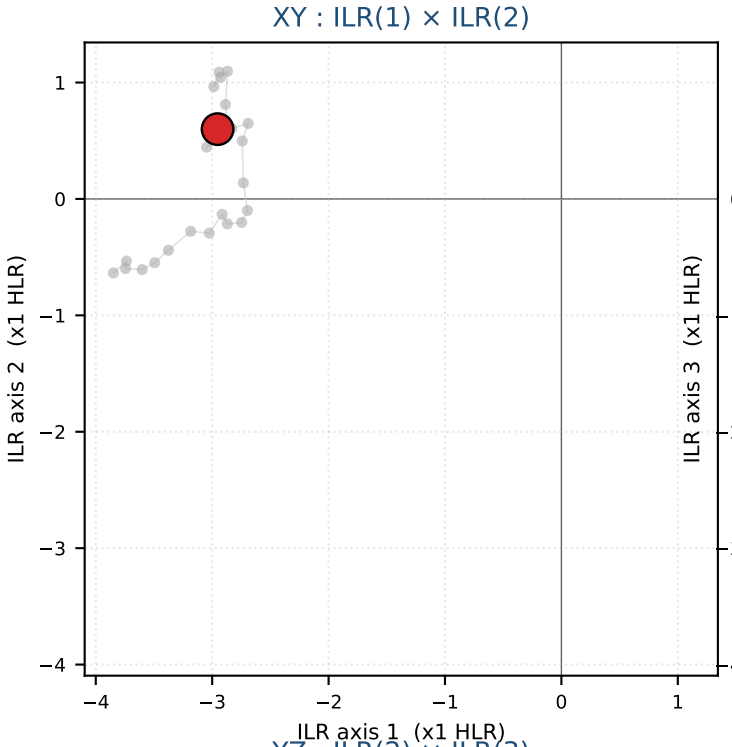
display factor: x1 (raw HLR x 1)

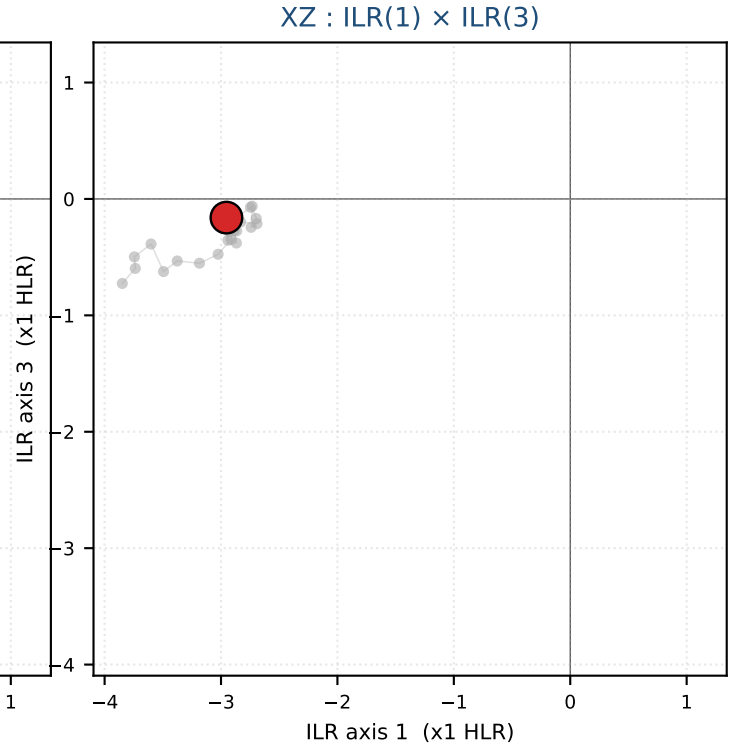
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=18 | 2018 | D=8 T=26 | level 1 first-principles

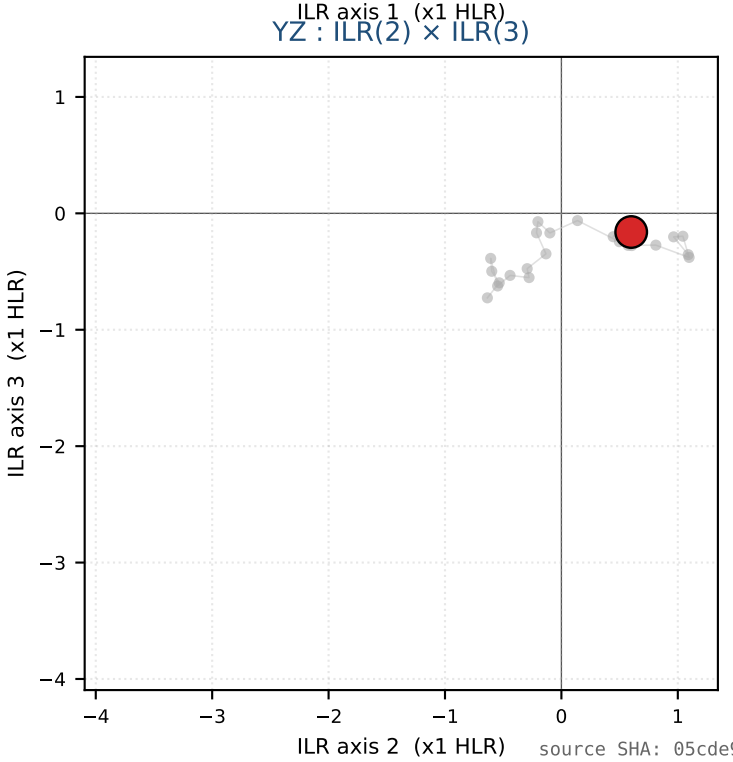
ember_ind | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

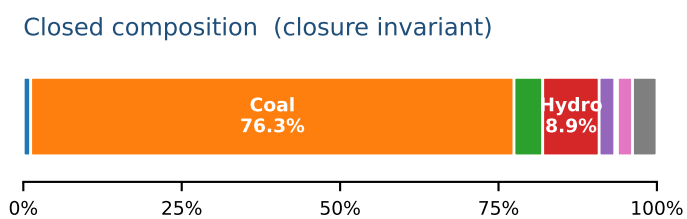




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0117	-1.144
Coal	0.7628	+3.033
Gas	0.0453	+0.210
Hydro	0.0890	+0.885
Nuclear	0.0248	-0.392
Other Fossil	0.0041	-2.200
Solar	0.0238	-0.434
Wind	0.0384	+0.043





closure: Σ = 1.000000 OK

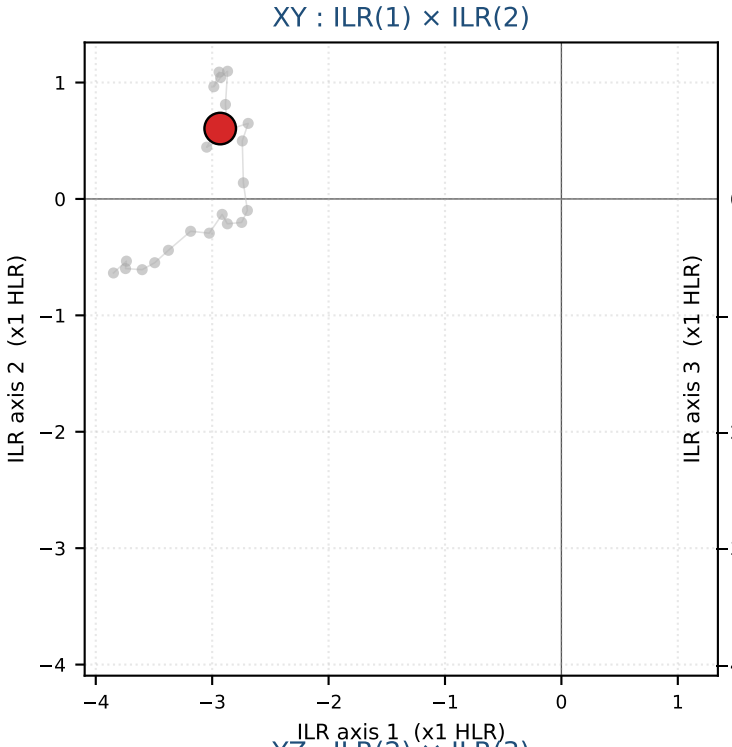
display factor: x1 (raw HLR x 1)

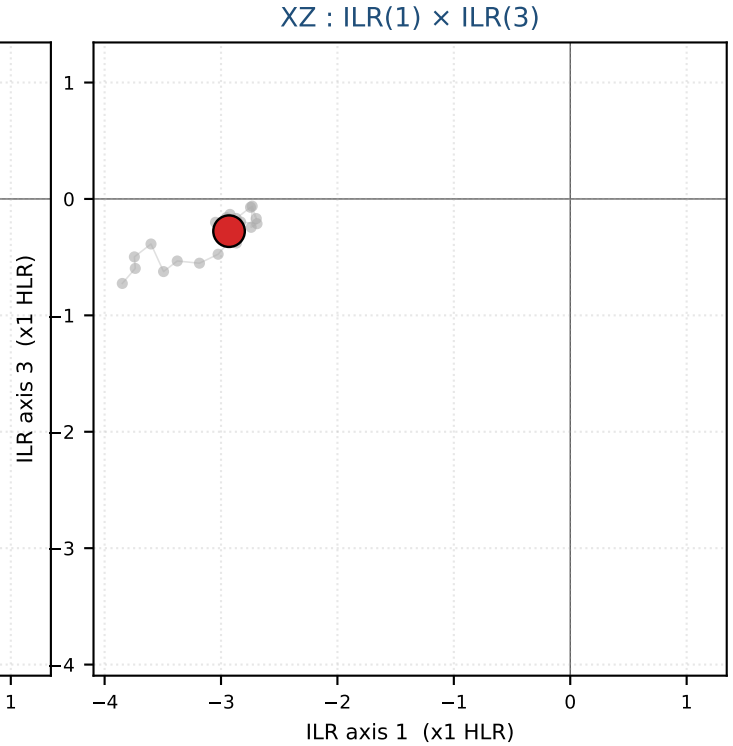
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=19 | 2019 | D=8 T=26 | level 1 first-principles

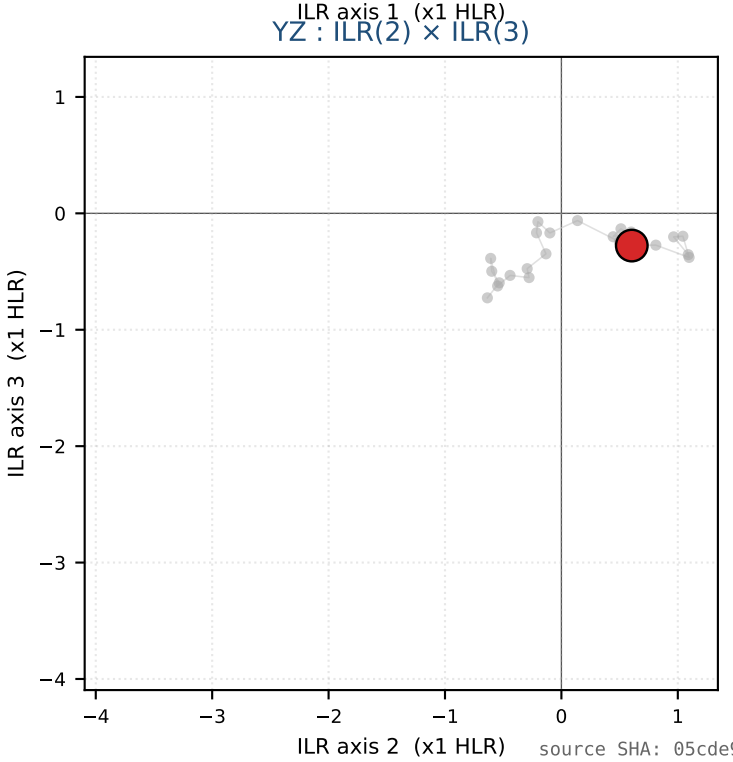
ember_ind | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

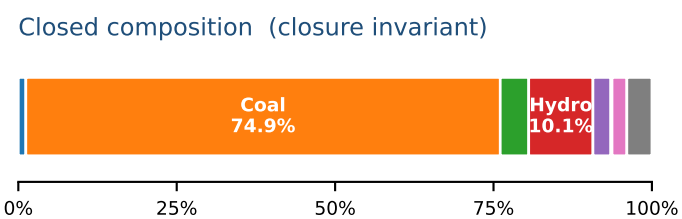




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0119	-1.069
Coal	0.7488	+3.076
Gas	0.0449	+0.263
Hydro	0.1014	+1.077
Nuclear	0.0282	-0.202
Other Fossil	0.0019	-2.891
Solar	0.0234	-0.390
Wind	0.0395	+0.135





closure: Σ = 1.000000 OK

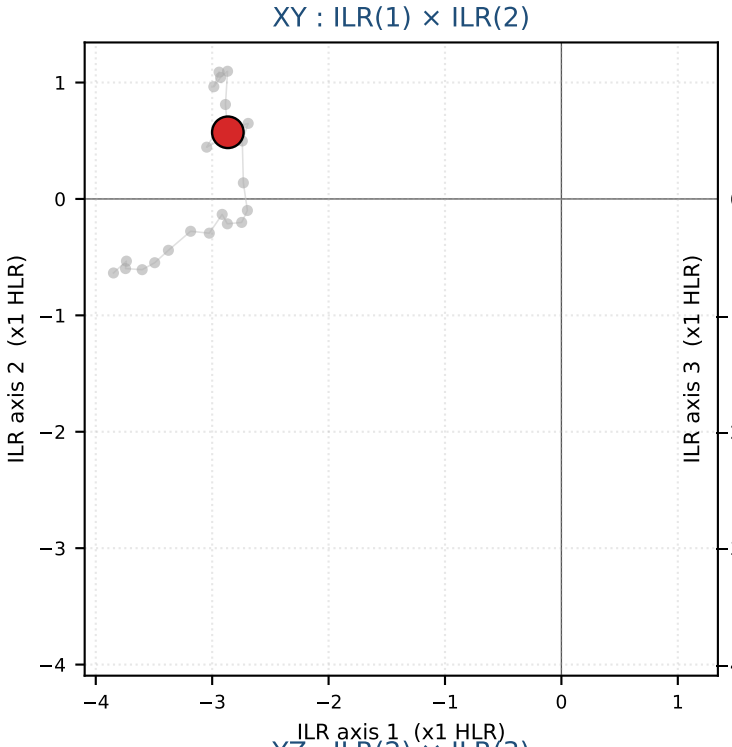
display factor: x1 (raw HLR x 1)

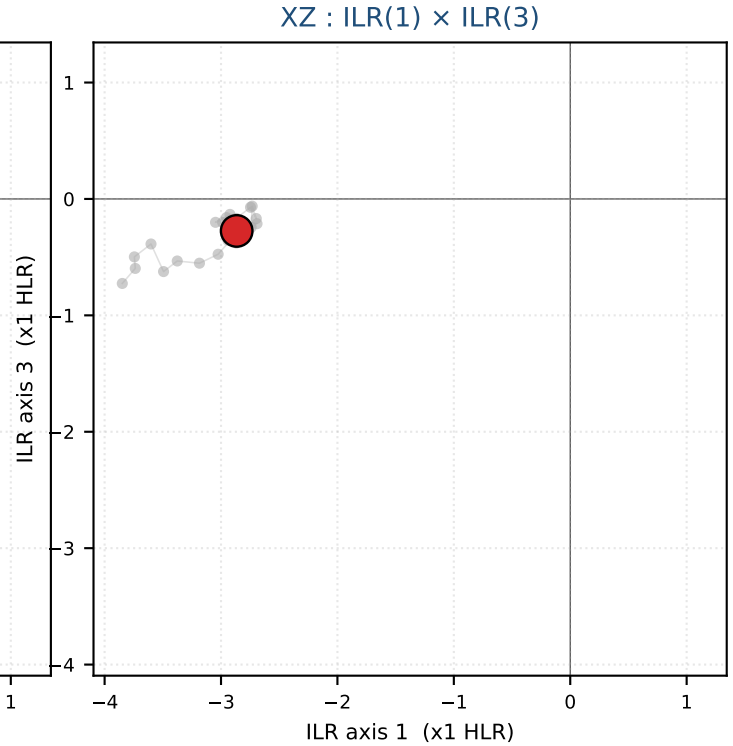
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=20 | 2020 | D=8 T=26 | level 1 first-principles

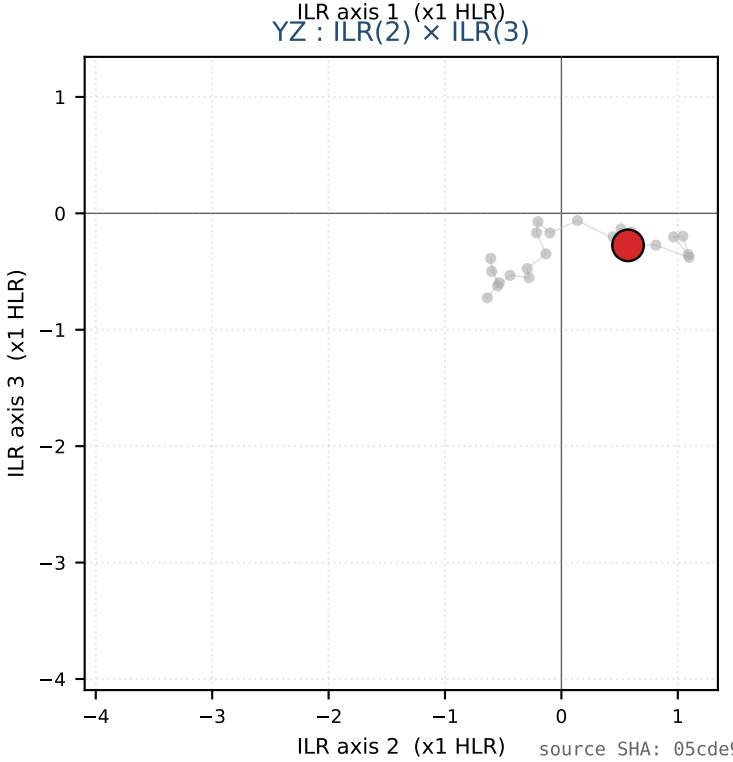
ember_ind | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

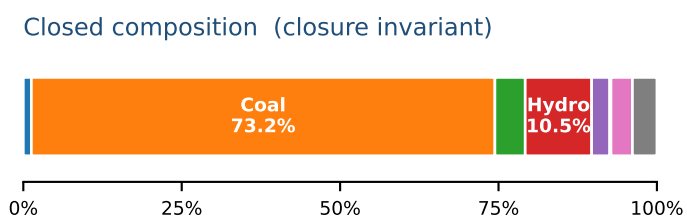




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0127	-1.066
Coal	0.7316	+2.987
Gas	0.0478	+0.259
Hydro	0.1049	+1.044
Nuclear	0.0285	-0.257
Other Fossil	0.0020	-2.927
Solar	0.0338	-0.087
Wind	0.0387	+0.046





closure: Σ = 1.000000 OK

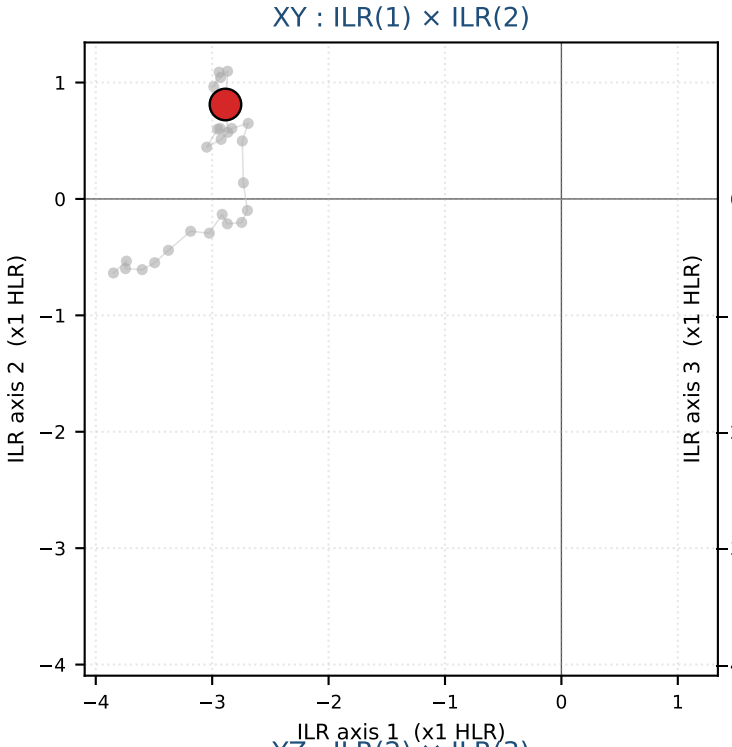
display factor: x1 (raw HLR x 1)

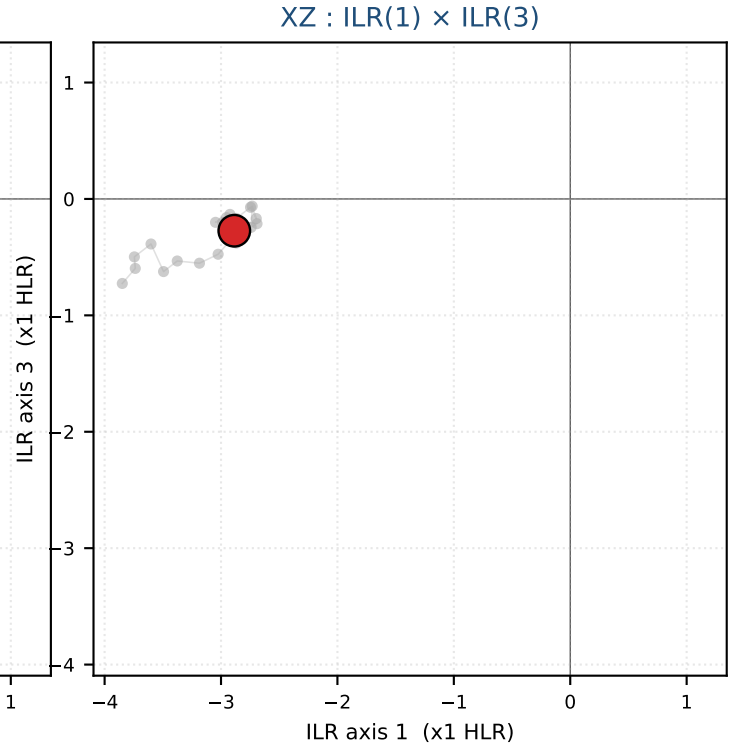
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=21 | 2021 | D=8 T=26 | level 1 first-principles

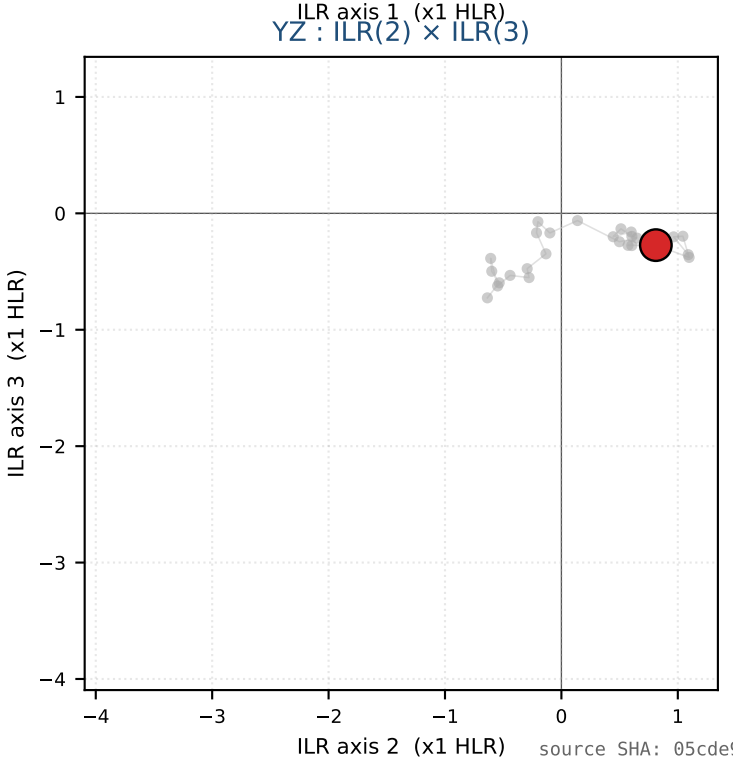
ember_ind | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

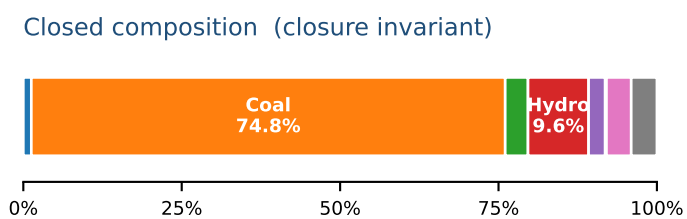




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0126	-1.043
Coal	0.7479	+3.038
Gas	0.0360	+0.004
Hydro	0.0957	+0.982
Nuclear	0.0262	-0.315
Other Fossil	0.0020	-2.876
Solar	0.0391	+0.086
Wind	0.0406	+0.124





closure: Σ = 1.000000 OK

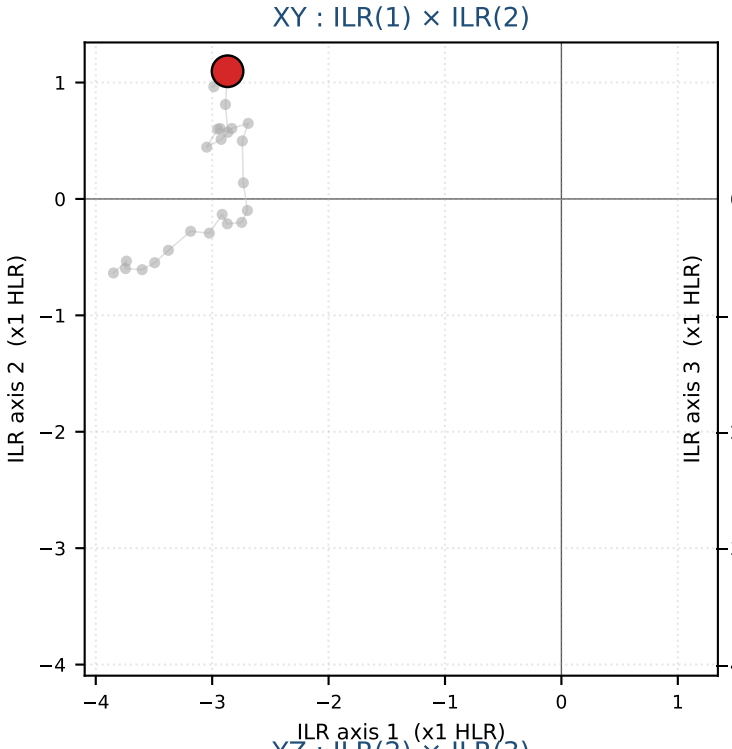
display factor: x1 (raw HLR x 1)

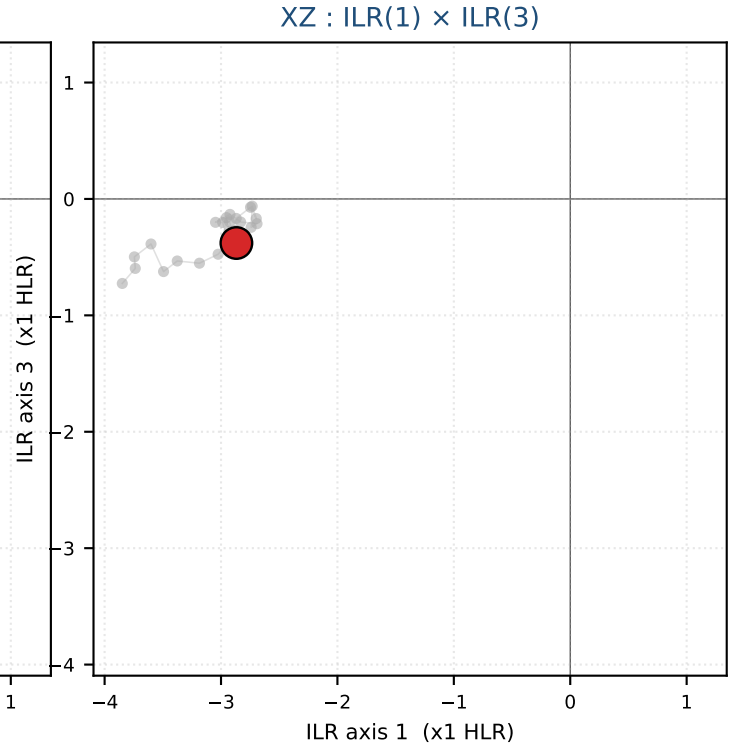
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=22 | 2022 | D=8 T=26 | level 1 first-principles

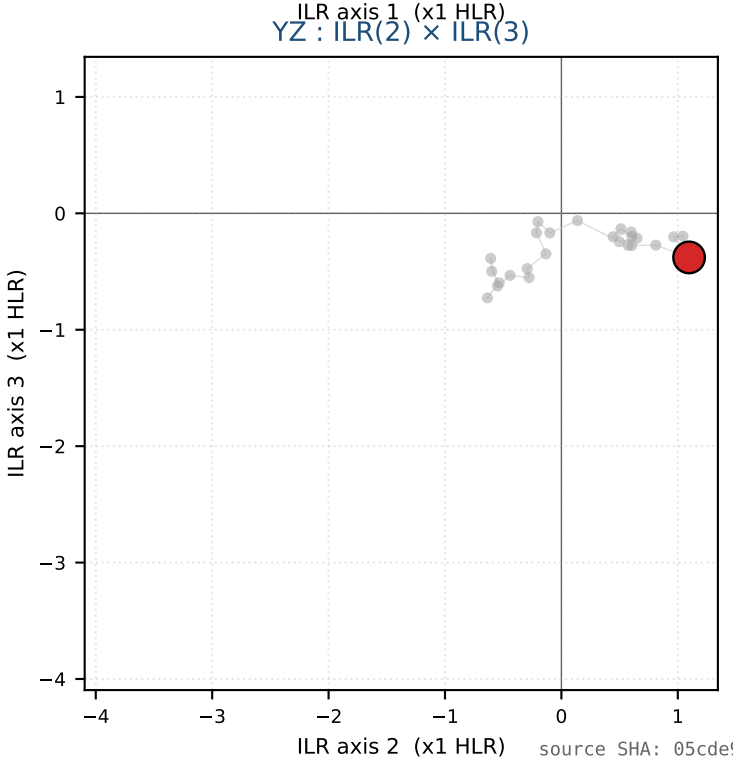
ember_ind | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

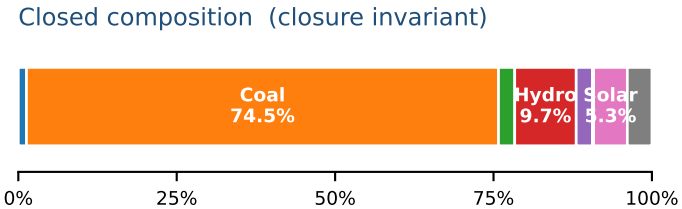




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0129	-1.007
Coal	0.7450	+3.048
Gas	0.0256	-0.323
Hydro	0.0970	+1.010
Nuclear	0.0256	-0.322
Other Fossil	0.0019	-2.908
Solar	0.0532	+0.408
Wind	0.0388	+0.094





closure: Σ = 1.000000 OK

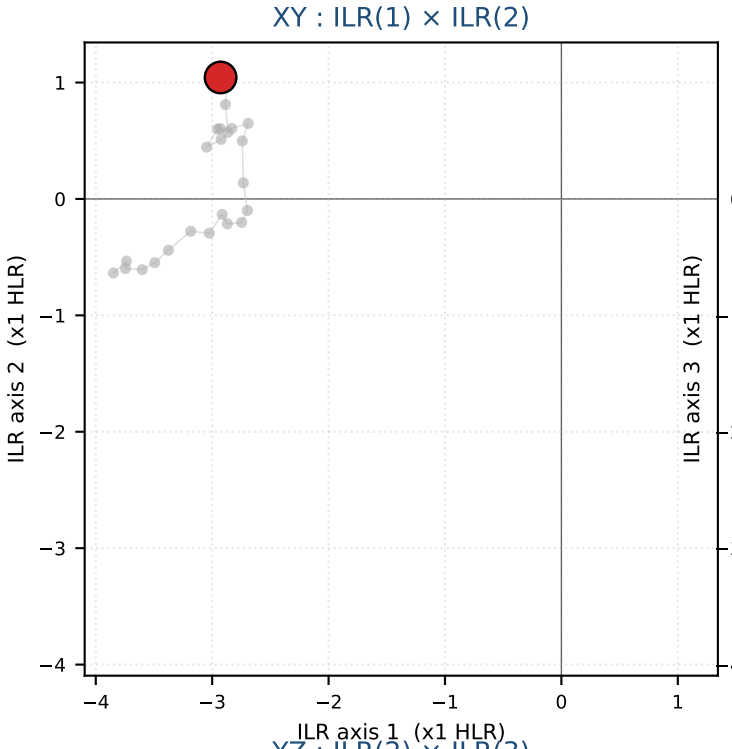
display factor: x1 (raw HLR x 1)

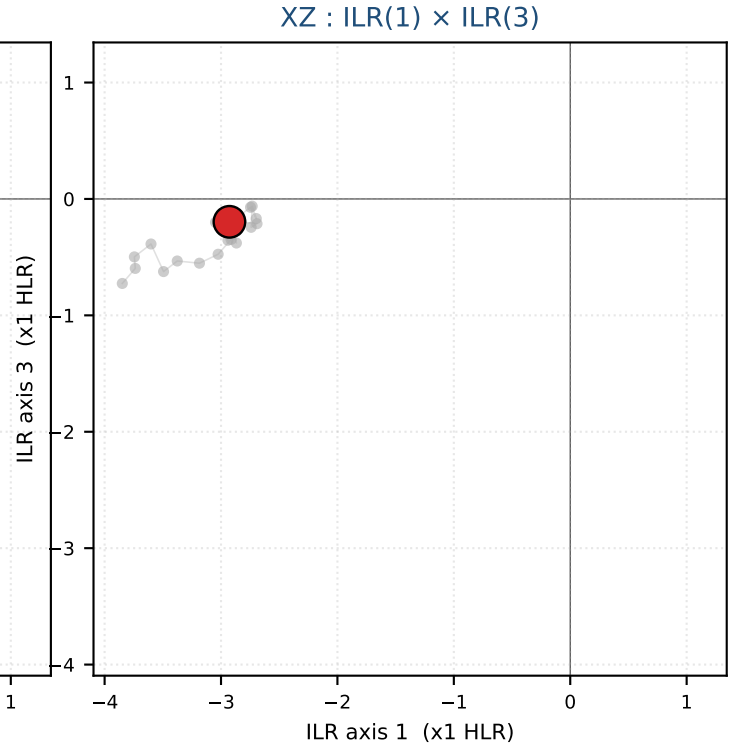
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=23 | 2023 | D=8 T=26 | level 1 first-principles

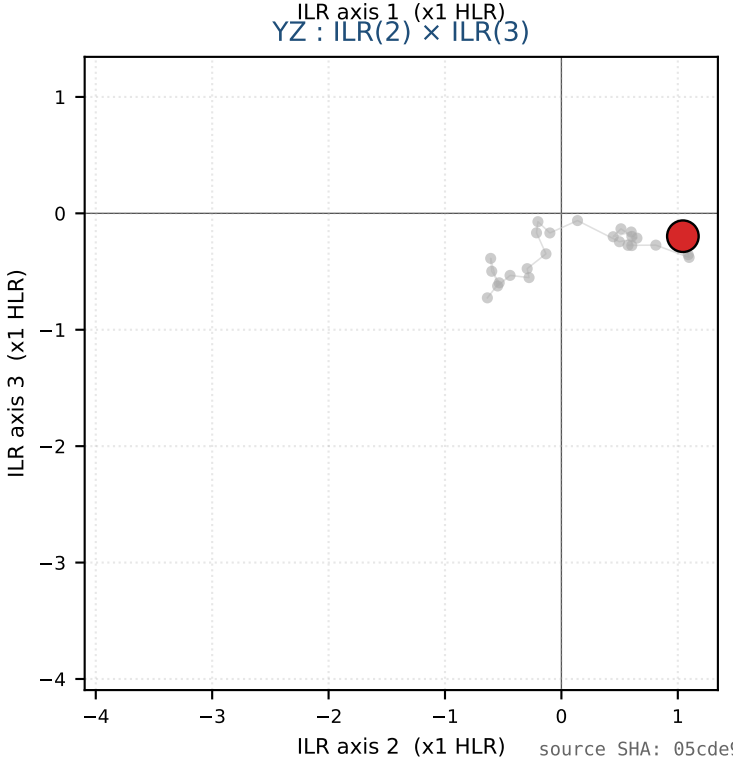
ember_ind | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

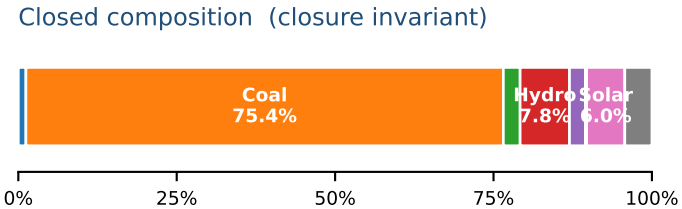




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0120	-1.080
Coal	0.7535	+3.060
Gas	0.0265	-0.288
Hydro	0.0779	+0.791
Nuclear	0.0251	-0.341
Other Fossil	0.0020	-2.868
Solar	0.0602	+0.533
Wind	0.0428	+0.192





closure: Σ = 1.000000 OK

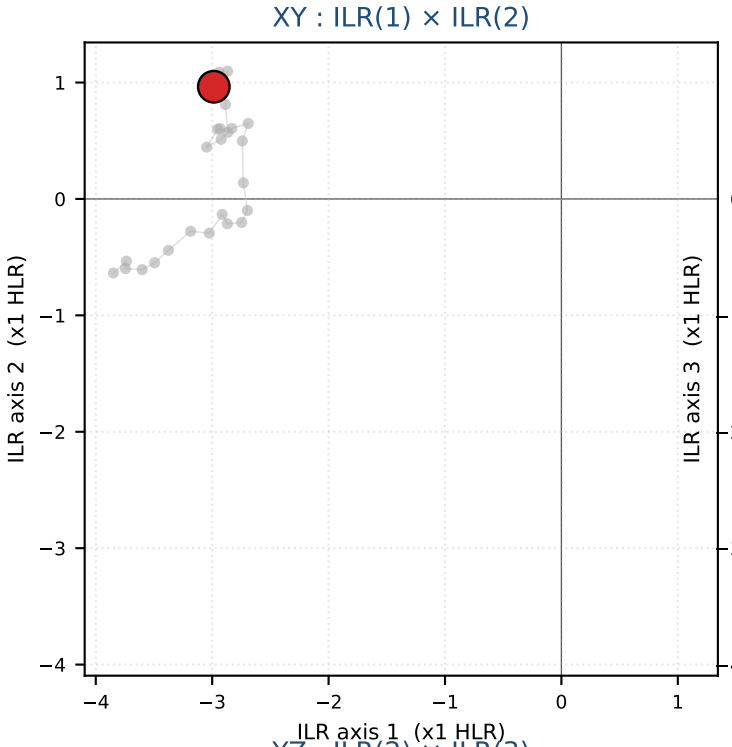
display factor: x1 (raw HLR x 1)

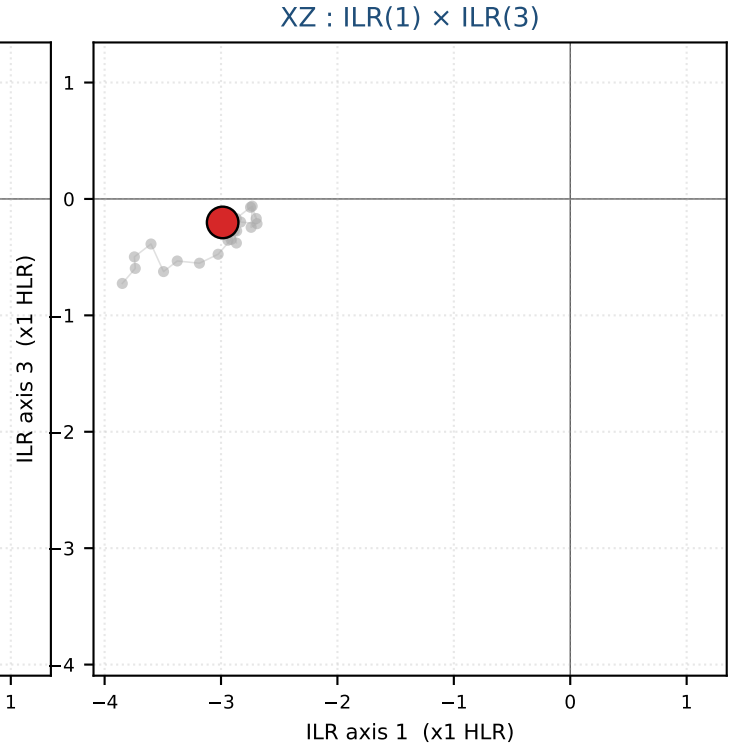
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=24 | 2024 | D=8 T=26 | level 1 first-principles

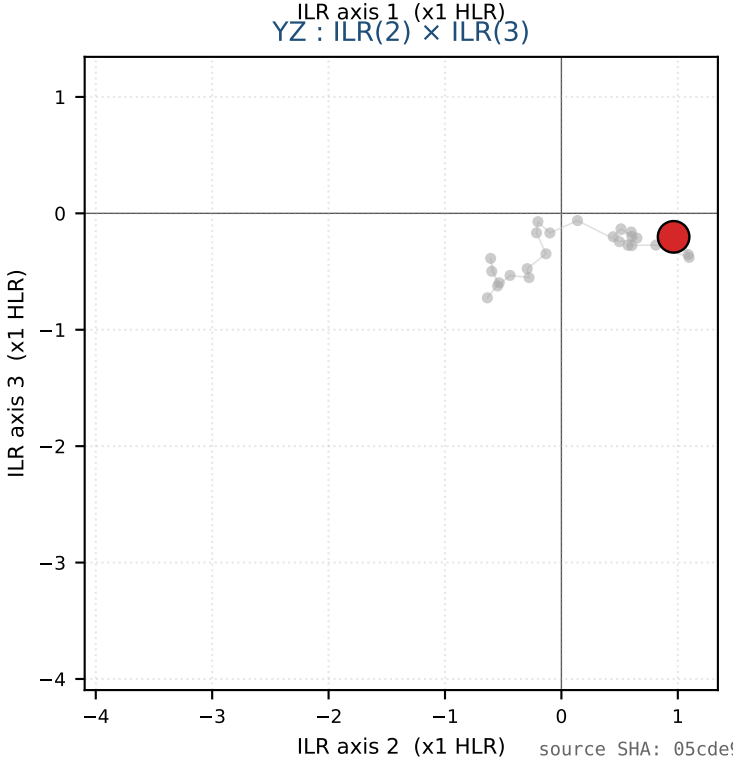
ember_ind | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

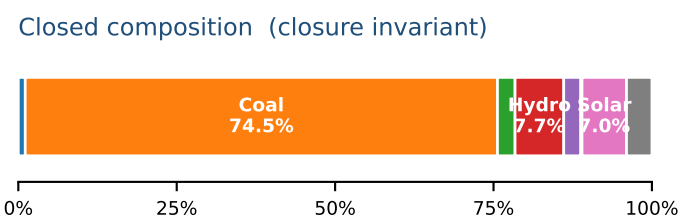




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0109	-1.184
Coal	0.7453	+3.039
Gas	0.0277	-0.253
Hydro	0.0769	+0.767
Nuclear	0.0269	-0.284
Other Fossil	0.0020	-2.878
Solar	0.0703	+0.677
Wind	0.0400	+0.115





closure: Σ = 1.000000 OK

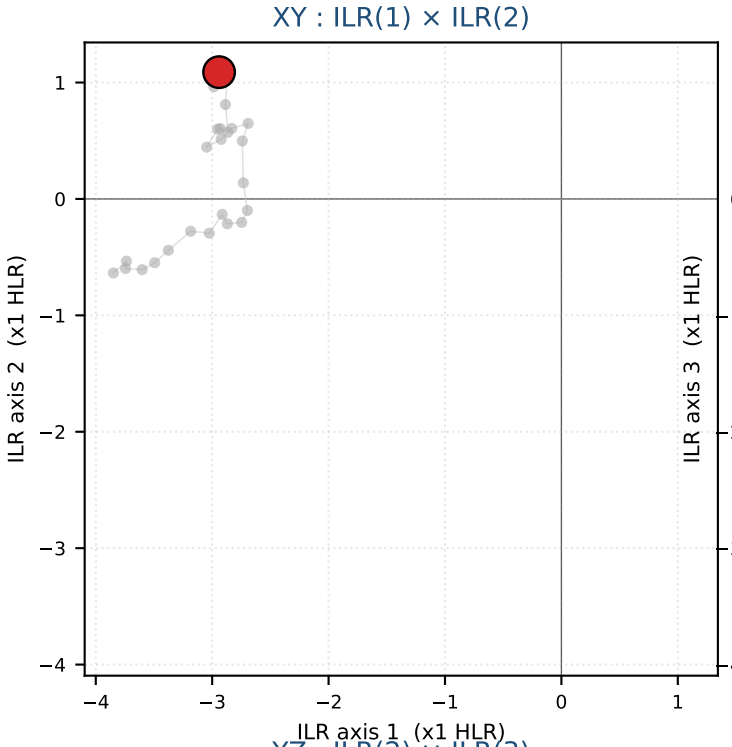
display factor: x1 (raw HLR x 1)

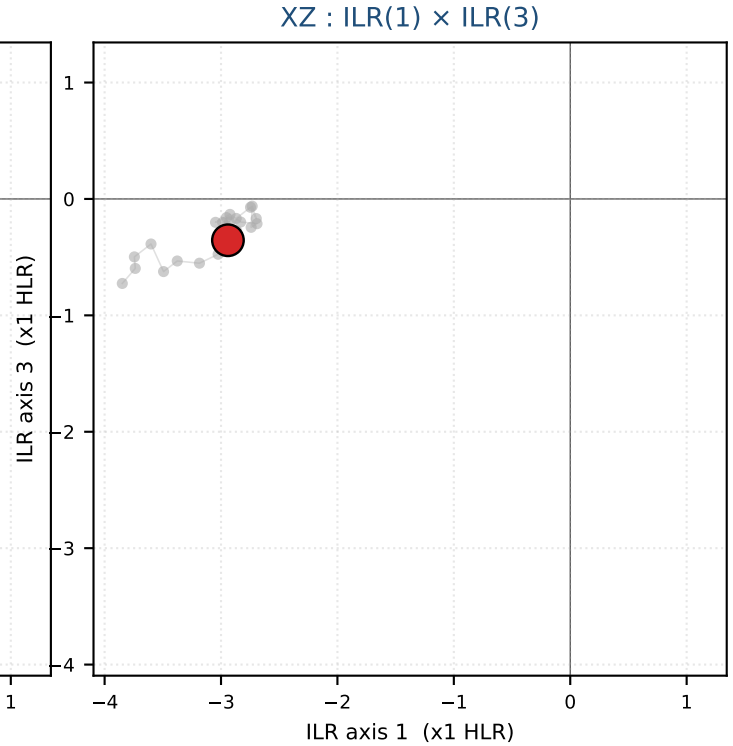
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=25 | 2025 | D=8 T=26 | level 1 first-principles

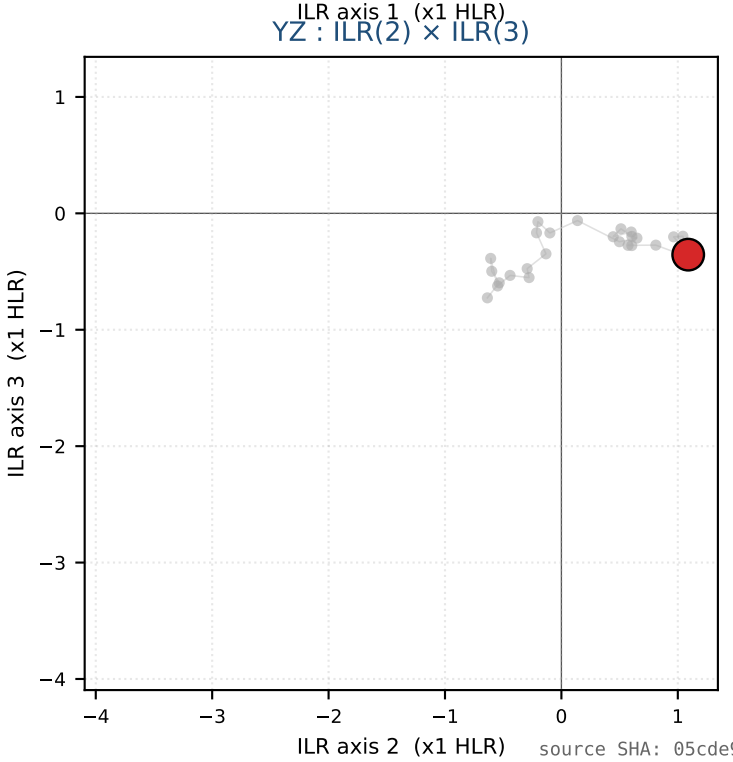
ember_ind | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

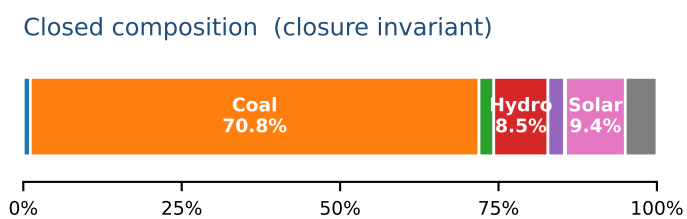




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0111	-1.215
Coal	0.7082	+2.944
Gas	0.0233	-0.470
Hydro	0.0855	+0.830
Nuclear	0.0259	-0.366
Other Fossil	0.0020	-2.941
Solar	0.0942	+0.926
Wind	0.0500	+0.292





closure: Σ = 1.000000 OK

display factor: x1 (raw HLR x 1)

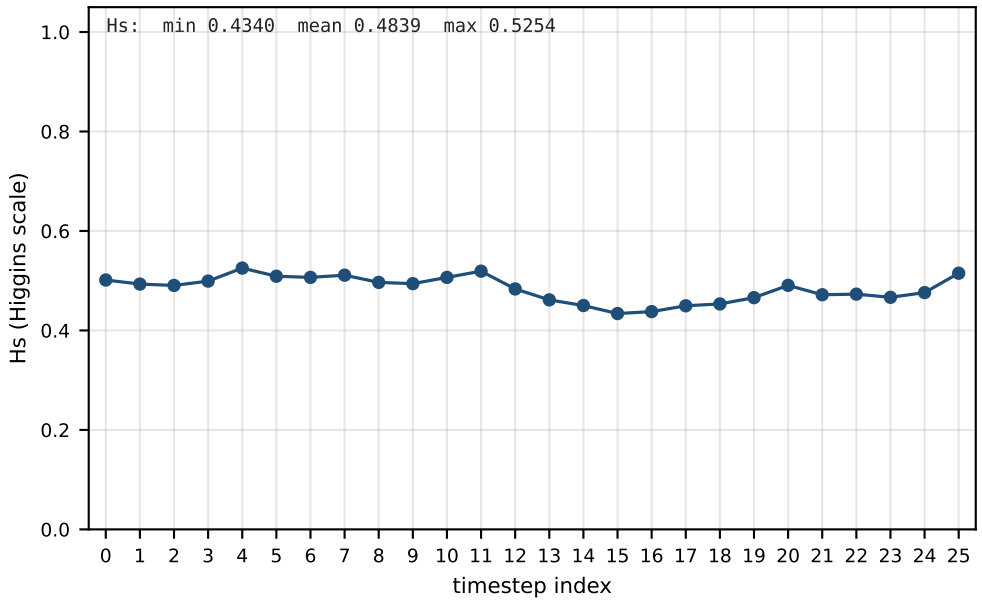
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

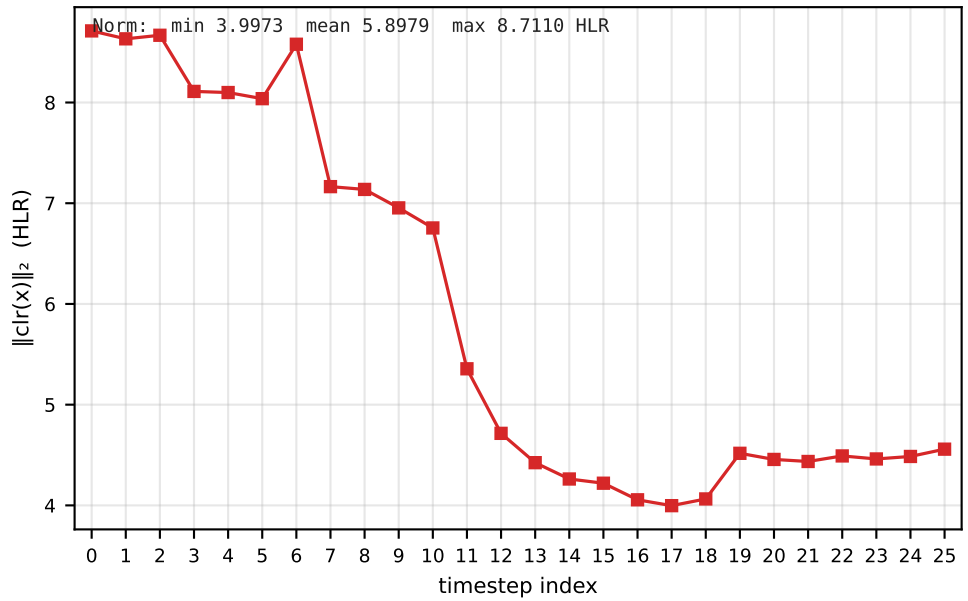
STAGE 1 SUMMARY | ember_ind | D=8 T=26 | Order 1 (first-principles closure)

Engine cnt 2.0.4 | Schema 2.1.0 | Generated 2026-05-06T01:24:14

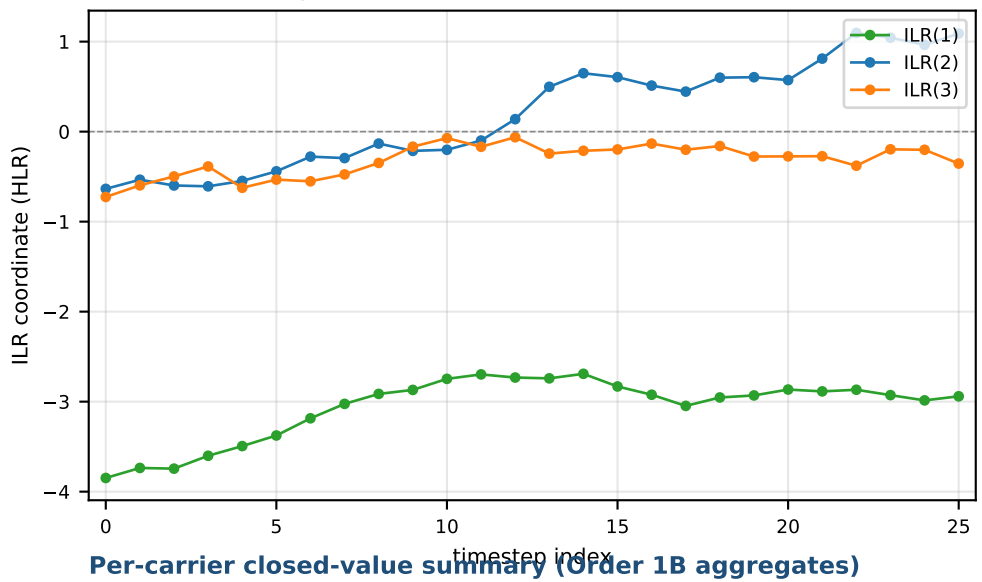
Hs trajectory — single-composition entropy / ln(D)



Aitchison norm — distance from centroid (HLR)



ILR axes 1–3 trajectories (Helmert basis)



Order-1 invariants (whole trajectory)

```
closure invariant     :  max |Σ closed − 1| = 4.44e-16
closure status        :  OK ✓

Hs range              :  [0.4340, 0.5254]
Aitchison norm range  :  [3.9973, 8.7110] HLR
ILR(1) range          :  [-3.8481, -2.6907]
ILR(2) range          :  [-0.6360, +1.0970]
ILR(3) range          :  [-0.7260, -0.0621]

Source file           :  ember_IND_India_generation_TWh.csv
Source SHA-256        :  05cde94fa8f37b98ae695073605eb97b...
Closed-data SHA-256   :  9d56e8be11bc49eb8e1a0c438d487c7f...
Content SHA-256       :  ad0da5d361bc14138f450b838ba2783e...
Ordering              :  temporal=True, method=by-time
Zero replacement δ    :  1e-15
```

Per-carrier closed-value summary (Order 1B aggregates)

#	carrier	min	mean	max	std
0	Bioenergy	0.0030	0.0105	0.0168	0.0040
1	Coal	0.6630	0.7185	0.7696	0.0344
2	Gas	0.0233	0.0743	0.1260	0.0345
3	Hydro	0.0769	0.1122	0.1540	0.0214
4	Nuclear	0.0184	0.0267	0.0322	0.0034
5	Other Fossil	0.0019	0.0165	0.0510	0.0166
6	Solar	0.0000	0.0168	0.0942	0.0256
7	Wind	0.0028	0.0245	0.0500	0.0141